Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:26:25

Measurement Details:

- **Sample:** Muestra9
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

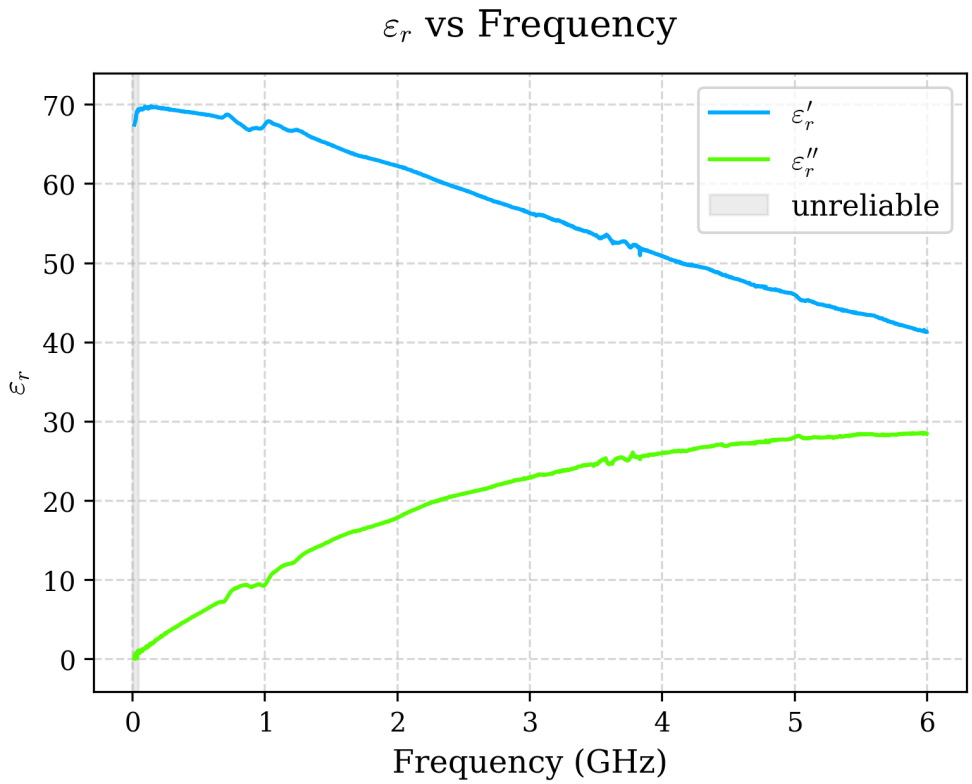
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	67.4456	0.0337	0.0005
0.020	67.7339	0.4292	0.0063
0.025	68.0571	0.7146	0.0105
0.030	68.7080	0.4955	0.0072
0.035	69.1299	0.0556	0.0008
0.040	69.0327	0.9782	0.0142
0.045	69.3972	1.0631	0.0153
0.050	69.3478	0.8217	0.0118
0.055	69.2814	0.7305	0.0105
0.060	69.4631	0.8914	0.0128
0.065	69.4896	1.1054	0.0159
0.070	69.4755	1.1081	0.0159
0.075	69.3324	0.9672	0.0139
0.080	69.4197	1.2797	0.0184
0.085	69.5124	1.2462	0.0179

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	69.4562	1.2644	0.0182
0.095	69.7346	1.2621	0.0181
0.100	69.5771	1.4102	0.0203
0.105	69.5764	1.5915	0.0229
0.110	69.6476	1.4005	0.0201
0.115	69.6489	1.6247	0.0233
0.120	69.4492	1.5195	0.0219
0.125	69.7008	1.6721	0.0240
0.130	69.5741	1.8617	0.0268
0.135	69.6636	1.7295	0.0248
0.140	69.7755	1.9727	0.0283
0.145	69.6007	1.9139	0.0275
0.150	69.6797	1.9807	0.0284
0.155	69.6808	1.9310	0.0277
0.160	69.6245	2.1117	0.0303
0.165	69.6605	2.1457	0.0308
0.170	69.6450	2.1662	0.0311
0.175	69.6541	2.3964	0.0344
0.180	69.6229	2.4126	0.0347
0.185	69.6794	2.3954	0.0344
0.190	69.5724	2.5285	0.0363
0.195	69.5612	2.5599	0.0368
0.200	69.5440	2.6103	0.0375
0.205	69.5952	2.6270	0.0377
0.210	69.5775	2.7691	0.0398
0.215	69.4798	2.7349	0.0394
0.220	69.5257	2.9327	0.0422
0.225	69.5058	2.8548	0.0411
0.230	69.4762	2.9268	0.0421
0.235	69.5491	2.9722	0.0427
0.240	69.5038	3.0764	0.0443
0.245	69.4527	3.1986	0.0461
0.250	69.4121	3.2875	0.0474
0.255	69.4557	3.1929	0.0460
0.260	69.4726	3.2967	0.0475
0.265	69.4141	3.3768	0.0486
0.270	69.3808	3.4430	0.0496
0.275	69.4584	3.4847	0.0502
0.280	69.3218	3.5369	0.0510
0.285	69.3901	3.5599	0.0513
0.290	69.3436	3.6068	0.0520
0.295	69.3410	3.6756	0.0530
0.300	69.3278	3.7768	0.0545
0.305	69.3029	3.8321	0.0553
0.310	69.2742	3.8652	0.0558
0.315	69.3237	3.8511	0.0556

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	69.2723	3.8928	0.0562
0.325	69.2570	4.0317	0.0582
0.330	69.2479	4.0489	0.0585
0.335	69.2425	4.0603	0.0586
0.340	69.2291	4.1335	0.0597
0.345	69.2308	4.1407	0.0598
0.350	69.1961	4.2532	0.0615
0.355	69.1960	4.3324	0.0626
0.360	69.1909	4.3488	0.0629
0.365	69.1659	4.3764	0.0633
0.370	69.1098	4.4415	0.0643
0.375	69.1196	4.5792	0.0662
0.380	69.1364	4.5554	0.0659
0.385	69.1334	4.5842	0.0663
0.390	69.1309	4.6436	0.0672
0.395	69.0544	4.6885	0.0679
0.400	69.0378	4.7323	0.0685
0.405	69.0639	4.8267	0.0699
0.410	69.0693	4.8254	0.0699
0.415	69.0435	4.8975	0.0709
0.420	69.0501	4.9259	0.0713
0.425	69.0650	4.9884	0.0722
0.430	69.0472	5.0544	0.0732
0.435	68.9801	5.0782	0.0736
0.440	69.0038	5.1281	0.0743
0.445	68.9825	5.2183	0.0756
0.450	68.9954	5.2761	0.0765
0.455	68.9459	5.2638	0.0763
0.460	68.9799	5.3467	0.0775
0.465	68.9543	5.3623	0.0778
0.470	68.9287	5.4549	0.0791
0.475	68.9281	5.4613	0.0792
0.480	68.9091	5.5143	0.0800
0.485	68.8925	5.5484	0.0805
0.490	68.8567	5.6225	0.0817
0.495	68.8691	5.6775	0.0824
0.500	68.8338	5.7016	0.0828
0.505	68.8329	5.7493	0.0835
0.510	68.8121	5.7959	0.0842
0.515	68.8053	5.8462	0.0850
0.520	68.8112	5.8692	0.0853
0.525	68.7610	5.9191	0.0861
0.530	68.7891	5.9992	0.0872
0.535	68.7764	6.0489	0.0879
0.540	68.7538	6.0946	0.0886
0.545	68.7608	6.1449	0.0894

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	68.7111	6.1562	0.0896
0.555	68.7286	6.2266	0.0906
0.560	68.6672	6.2754	0.0914
0.565	68.6835	6.3277	0.0921
0.570	68.6791	6.3845	0.0930
0.575	68.7006	6.4143	0.0934
0.580	68.6760	6.4896	0.0945
0.585	68.6381	6.5167	0.0949
0.590	68.6169	6.5592	0.0956
0.595	68.6023	6.6386	0.0968
0.600	68.6116	6.6834	0.0974
0.605	68.5568	6.7208	0.0980
0.610	68.5538	6.7489	0.0984
0.615	68.5453	6.8214	0.0995
0.620	68.5154	6.8359	0.0998
0.625	68.5032	6.9079	0.1008
0.630	68.4956	6.9130	0.1009
0.635	68.4518	7.0067	0.1024
0.640	68.4482	7.0233	0.1026
0.645	68.3923	7.0803	0.1035
0.650	68.3891	7.1131	0.1040
0.655	68.3620	7.1571	0.1047
0.660	68.3564	7.1470	0.1046
0.665	68.3592	7.1711	0.1049
0.670	68.3398	7.2042	0.1054
0.675	68.3400	7.1781	0.1050
0.680	68.3268	7.2097	0.1055
0.685	68.3783	7.1958	0.1052
0.690	68.4390	7.2046	0.1053
0.695	68.4901	7.2513	0.1059
0.700	68.5542	7.3549	0.1073
0.705	68.6640	7.4906	0.1091
0.710	68.7001	7.5602	0.1100
0.715	68.7149	7.7350	0.1126
0.720	68.7297	7.8851	0.1147
0.725	68.6843	8.0169	0.1167
0.730	68.6507	8.1306	0.1184
0.735	68.5718	8.2649	0.1205
0.740	68.5140	8.4113	0.1228
0.745	68.4530	8.4901	0.1240
0.750	68.3408	8.5999	0.1258
0.755	68.2792	8.6612	0.1268
0.760	68.2260	8.7214	0.1278
0.765	68.1359	8.7979	0.1291
0.770	68.0488	8.8618	0.1302
0.775	67.9925	8.8449	0.1301

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	67.9144	8.9283	0.1315
0.785	67.8328	8.9661	0.1322
0.790	67.7771	8.9561	0.1321
0.795	67.7111	9.0095	0.1331
0.800	67.6497	9.0323	0.1335
0.805	67.6133	9.1145	0.1348
0.810	67.5505	9.1433	0.1354
0.815	67.5455	9.1889	0.1360
0.820	67.4659	9.2113	0.1365
0.825	67.3957	9.2154	0.1367
0.830	67.3549	9.2626	0.1375
0.835	67.2784	9.3059	0.1383
0.840	67.1938	9.2889	0.1382
0.845	67.1762	9.3021	0.1385
0.850	67.0820	9.3274	0.1390
0.855	67.0226	9.3566	0.1396
0.860	66.9269	9.3177	0.1392
0.865	66.8886	9.3617	0.1400
0.870	66.8666	9.3267	0.1395
0.875	66.8281	9.2443	0.1383
0.880	66.7474	9.2007	0.1378
0.885	66.7427	9.1546	0.1372
0.890	66.8100	9.1467	0.1369
0.895	66.8119	9.0789	0.1359
0.900	66.8899	9.0404	0.1352
0.905	66.9570	9.0956	0.1358
0.910	66.9762	9.1346	0.1364
0.915	66.9597	9.1914	0.1373
0.920	66.9704	9.1926	0.1373
0.925	67.0238	9.2432	0.1379
0.930	67.0301	9.2606	0.1382
0.935	67.0182	9.3150	0.1390
0.940	67.0451	9.3702	0.1398
0.945	67.0017	9.4031	0.1403
0.950	66.9689	9.4047	0.1404
0.955	66.9472	9.4329	0.1409
0.960	66.9308	9.3914	0.1403
0.965	66.9514	9.4074	0.1405
0.970	66.9562	9.3635	0.1398
0.975	66.9654	9.3076	0.1390
0.980	67.0352	9.2817	0.1385
0.985	67.1197	9.2303	0.1375
0.990	67.1971	9.2705	0.1380
0.995	67.3121	9.2632	0.1376
1.000	67.4121	9.2950	0.1379
1.005	67.5618	9.4284	0.1396

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	67.6709	9.5431	0.1410
1.015	67.7137	9.6523	0.1425
1.020	67.8149	9.8116	0.1447
1.025	67.8838	9.9371	0.1464
1.030	67.8748	10.1029	0.1488
1.035	67.8738	10.2345	0.1508
1.040	67.8457	10.3850	0.1531
1.045	67.7825	10.5133	0.1551
1.050	67.7472	10.6409	0.1571
1.055	67.6236	10.7177	0.1585
1.060	67.6407	10.8308	0.1601
1.065	67.5425	10.8448	0.1606
1.070	67.5473	10.9268	0.1618
1.075	67.4912	11.0112	0.1632
1.080	67.4632	11.0363	0.1636
1.085	67.4365	11.0660	0.1641
1.090	67.4342	11.1585	0.1655
1.095	67.4079	11.2493	0.1669
1.100	67.3831	11.3087	0.1678
1.105	67.3384	11.3654	0.1688
1.110	67.2925	11.4567	0.1703
1.115	67.2715	11.5126	0.1711
1.120	67.2289	11.5515	0.1718
1.125	67.1942	11.6057	0.1727
1.130	67.1264	11.6956	0.1742
1.135	67.0595	11.7341	0.1750
1.140	66.9868	11.7350	0.1752
1.145	66.9673	11.8168	0.1765
1.150	66.9373	11.8000	0.1763
1.155	66.8969	11.8873	0.1777
1.160	66.8259	11.8855	0.1779
1.165	66.8243	11.9244	0.1784
1.170	66.7764	11.9143	0.1784
1.175	66.7100	11.9792	0.1796
1.180	66.7220	12.0062	0.1799
1.185	66.7001	11.9909	0.1798
1.190	66.6549	11.9934	0.1799
1.195	66.6800	12.0380	0.1805
1.200	66.7030	12.0683	0.1809
1.205	66.6373	12.0628	0.1810
1.210	66.6822	12.0905	0.1813
1.215	66.6872	12.1510	0.1822
1.220	66.7375	12.1659	0.1823
1.225	66.7393	12.2158	0.1830
1.230	66.7826	12.3293	0.1846
1.235	66.7374	12.3833	0.1856

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	66.7987	12.4436	0.1863
1.245	66.7965	12.5336	0.1876
1.250	66.7233	12.6253	0.1892
1.255	66.7128	12.7013	0.1904
1.260	66.6922	12.7570	0.1913
1.265	66.6502	12.8738	0.1932
1.270	66.6227	12.9137	0.1938
1.275	66.5874	12.9969	0.1952
1.280	66.5420	13.0669	0.1964
1.285	66.5497	13.1339	0.1974
1.290	66.4303	13.1771	0.1984
1.295	66.3833	13.2611	0.1998
1.300	66.3526	13.3103	0.2006
1.305	66.3174	13.3604	0.2015
1.310	66.2677	13.4234	0.2026
1.315	66.2336	13.4643	0.2033
1.320	66.1433	13.4909	0.2040
1.325	66.0881	13.5614	0.2052
1.330	66.0506	13.6105	0.2061
1.335	65.9990	13.6455	0.2068
1.340	65.9923	13.6667	0.2071
1.345	65.9120	13.7205	0.2082
1.350	65.8790	13.7890	0.2093
1.355	65.8075	13.7628	0.2091
1.360	65.8300	13.8230	0.2100
1.365	65.7806	13.8676	0.2108
1.370	65.7341	13.9035	0.2115
1.375	65.7355	13.9278	0.2119
1.380	65.6739	13.9957	0.2131
1.385	65.6522	14.0516	0.2140
1.390	65.6033	14.0608	0.2143
1.395	65.5811	14.1216	0.2153
1.400	65.5685	14.1394	0.2156
1.405	65.4924	14.2054	0.2169
1.410	65.4920	14.2274	0.2172
1.415	65.4175	14.2321	0.2176
1.420	65.4158	14.2903	0.2185
1.425	65.3643	14.2805	0.2185
1.430	65.3147	14.3720	0.2200
1.435	65.2838	14.4008	0.2206
1.440	65.2652	14.4358	0.2212
1.445	65.2424	14.4808	0.2220
1.450	65.2452	14.5073	0.2224
1.455	65.1924	14.5655	0.2234
1.460	65.1711	14.6352	0.2246
1.465	65.1418	14.6633	0.2251

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	65.0885	14.6956	0.2258
1.475	65.0197	14.6735	0.2257
1.480	65.0523	14.7456	0.2267
1.485	64.9931	14.7821	0.2274
1.490	64.9840	14.8183	0.2280
1.495	64.9161	14.8741	0.2291
1.500	64.8885	14.9431	0.2303
1.505	64.8498	14.9793	0.2310
1.510	64.7928	14.9912	0.2314
1.515	64.8408	15.0683	0.2324
1.520	64.7493	15.1110	0.2334
1.525	64.7338	15.1649	0.2343
1.530	64.6810	15.1646	0.2345
1.535	64.6477	15.1906	0.2350
1.540	64.6082	15.2358	0.2358
1.545	64.5759	15.3030	0.2370
1.550	64.5581	15.3117	0.2372
1.555	64.5048	15.3226	0.2375
1.560	64.4879	15.3990	0.2388
1.565	64.4434	15.4578	0.2399
1.570	64.3856	15.4724	0.2403
1.575	64.3710	15.5075	0.2409
1.580	64.3569	15.4959	0.2408
1.585	64.2821	15.5530	0.2419
1.590	64.2672	15.5973	0.2427
1.595	64.2239	15.6260	0.2433
1.600	64.2080	15.6510	0.2438
1.605	64.1730	15.6796	0.2443
1.610	64.1285	15.6982	0.2448
1.615	64.0909	15.7478	0.2457
1.620	64.0784	15.8071	0.2467
1.625	64.0625	15.8524	0.2475
1.630	63.9784	15.8421	0.2476
1.635	63.9764	15.8994	0.2485
1.640	63.9665	15.9119	0.2488
1.645	63.9054	15.9479	0.2496
1.650	63.8251	15.9455	0.2498
1.655	63.8059	15.9637	0.2502
1.660	63.7839	16.0576	0.2518
1.665	63.8486	16.1007	0.2522
1.670	63.7508	16.1093	0.2527
1.675	63.6949	16.1161	0.2530
1.680	63.7175	16.1263	0.2531
1.685	63.6406	16.0796	0.2527
1.690	63.6446	16.1540	0.2538
1.695	63.6278	16.1514	0.2538

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	63.5765	16.1814	0.2545
1.705	63.5344	16.2139	0.2552
1.710	63.5451	16.2122	0.2551
1.715	63.5064	16.1883	0.2549
1.720	63.4794	16.2916	0.2566
1.725	63.4462	16.2896	0.2567
1.730	63.4936	16.3143	0.2569
1.735	63.4255	16.3620	0.2580
1.740	63.4055	16.3775	0.2583
1.745	63.3721	16.4478	0.2595
1.750	63.3573	16.3790	0.2585
1.755	63.3834	16.4534	0.2596
1.760	63.3482	16.4568	0.2598
1.765	63.3661	16.4894	0.2602
1.770	63.2957	16.5345	0.2612
1.775	63.3109	16.5941	0.2621
1.780	63.2587	16.5865	0.2622
1.785	63.2055	16.6308	0.2631
1.790	63.1831	16.6198	0.2630
1.795	63.1940	16.6546	0.2635
1.800	63.1577	16.6487	0.2636
1.805	63.1301	16.6972	0.2645
1.810	63.1144	16.7191	0.2649
1.815	63.1072	16.7215	0.2650
1.820	63.0860	16.7527	0.2656
1.825	63.0795	16.7884	0.2661
1.830	63.0138	16.8361	0.2672
1.835	63.0123	16.8633	0.2676
1.840	62.9904	16.9056	0.2684
1.845	62.9799	16.9270	0.2688
1.850	62.9543	16.9411	0.2691
1.855	62.9287	16.9595	0.2695
1.860	62.9020	16.9665	0.2697
1.865	62.8920	17.0050	0.2704
1.870	62.8909	17.0208	0.2706
1.875	62.8378	17.0812	0.2718
1.880	62.8203	17.0790	0.2719
1.885	62.7604	17.1199	0.2728
1.890	62.7616	17.1352	0.2730
1.895	62.7231	17.1492	0.2734
1.900	62.7200	17.2292	0.2747
1.905	62.6871	17.2220	0.2747
1.910	62.6483	17.2324	0.2751
1.915	62.6291	17.2999	0.2762
1.920	62.6114	17.2910	0.2762
1.925	62.6136	17.3269	0.2767

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	62.5637	17.3172	0.2768
1.935	62.5645	17.3737	0.2777
1.940	62.5017	17.3558	0.2777
1.945	62.4871	17.4687	0.2796
1.950	62.4613	17.4857	0.2799
1.955	62.4508	17.5225	0.2806
1.960	62.4396	17.5095	0.2804
1.965	62.3964	17.5691	0.2816
1.970	62.3847	17.5760	0.2817
1.975	62.3894	17.6624	0.2831
1.980	62.3177	17.6607	0.2834
1.985	62.3322	17.6545	0.2832
1.990	62.2980	17.7289	0.2846
1.995	62.2726	17.7708	0.2854
2.000	62.2263	17.8446	0.2868
2.005	62.2387	17.8402	0.2866
2.010	62.2195	17.9347	0.2882
2.015	62.1441	17.9264	0.2885
2.020	62.1247	17.9727	0.2893
2.025	62.1265	18.0321	0.2902
2.030	62.1230	18.0284	0.2902
2.035	62.1009	18.1019	0.2915
2.040	62.0711	18.1308	0.2921
2.045	61.9721	18.1867	0.2935
2.050	62.0501	18.2732	0.2945
2.055	62.0036	18.2760	0.2948
2.060	61.9404	18.3410	0.2961
2.065	61.9689	18.3546	0.2962
2.070	61.8802	18.3928	0.2972
2.075	61.8290	18.4231	0.2980
2.080	61.8022	18.4828	0.2991
2.085	61.7925	18.5134	0.2996
2.090	61.7744	18.5507	0.3003
2.095	61.7337	18.5713	0.3008
2.100	61.7005	18.6681	0.3026
2.105	61.6770	18.6718	0.3027
2.110	61.6825	18.6975	0.3031
2.115	61.6020	18.7457	0.3043
2.120	61.5547	18.7611	0.3048
2.125	61.5329	18.8171	0.3058
2.130	61.4894	18.8345	0.3063
2.135	61.4963	18.8632	0.3067
2.140	61.4636	18.9092	0.3076
2.145	61.4015	18.9385	0.3084
2.150	61.4170	18.9517	0.3086
2.155	61.3486	19.0048	0.3098

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	61.2978	19.0562	0.3109
2.165	61.3260	19.0785	0.3111
2.170	61.2584	19.1220	0.3122
2.175	61.2581	19.1676	0.3129
2.180	61.2149	19.1967	0.3136
2.185	61.1949	19.2287	0.3142
2.190	61.1394	19.2747	0.3153
2.195	61.0639	19.3182	0.3164
2.200	61.0561	19.3588	0.3171
2.205	61.0142	19.3390	0.3170
2.210	61.0298	19.4364	0.3185
2.215	60.9636	19.4482	0.3190
2.220	60.9552	19.4762	0.3195
2.225	60.9215	19.5364	0.3207
2.230	60.8810	19.5678	0.3214
2.235	60.8715	19.5903	0.3218
2.240	60.8209	19.5777	0.3219
2.245	60.7594	19.6704	0.3237
2.250	60.7487	19.6703	0.3238
2.255	60.7153	19.6855	0.3242
2.260	60.6700	19.7655	0.3258
2.265	60.6537	19.7927	0.3263
2.270	60.5989	19.7697	0.3262
2.275	60.5351	19.8200	0.3274
2.280	60.5180	19.8298	0.3277
2.285	60.5214	19.9018	0.3288
2.290	60.4628	19.8990	0.3291
2.295	60.4658	19.8776	0.3287
2.300	60.3817	19.8805	0.3292
2.305	60.3548	20.0213	0.3317
2.310	60.2999	19.9386	0.3307
2.315	60.3337	19.9935	0.3314
2.320	60.2844	20.0461	0.3325
2.325	60.1807	20.0315	0.3329
2.330	60.2443	20.1478	0.3344
2.335	60.1670	20.1342	0.3346
2.340	60.1723	20.0963	0.3340
2.345	60.0957	20.1937	0.3360
2.350	60.0850	20.2263	0.3366
2.355	60.0195	20.1918	0.3364
2.360	60.0364	20.2540	0.3374
2.365	60.0119	20.3000	0.3383
2.370	59.9727	20.2758	0.3381
2.375	59.9375	20.2646	0.3381
2.380	59.9231	20.3495	0.3396
2.385	59.8652	20.3901	0.3406

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	59.8818	20.3889	0.3405
2.395	59.8291	20.4740	0.3422
2.400	59.7843	20.4869	0.3427
2.405	59.7358	20.4424	0.3422
2.410	59.7378	20.4751	0.3427
2.415	59.6830	20.5170	0.3438
2.420	59.6882	20.5707	0.3446
2.425	59.6310	20.5589	0.3448
2.430	59.6083	20.5569	0.3449
2.435	59.6011	20.5652	0.3450
2.440	59.5504	20.6078	0.3461
2.445	59.5352	20.6240	0.3464
2.450	59.5406	20.6336	0.3465
2.455	59.4673	20.6515	0.3473
2.460	59.4541	20.6805	0.3478
2.465	59.4520	20.7207	0.3485
2.470	59.3749	20.7227	0.3490
2.475	59.3667	20.7448	0.3494
2.480	59.3258	20.7656	0.3500
2.485	59.3467	20.7966	0.3504
2.490	59.2905	20.8006	0.3508
2.495	59.2484	20.8563	0.3520
2.500	59.2387	20.8314	0.3517
2.505	59.1962	20.8519	0.3523
2.510	59.1903	20.8838	0.3528
2.515	59.1500	20.8958	0.3533
2.520	59.1249	20.9332	0.3541
2.525	59.0515	20.9627	0.3550
2.530	59.0490	20.9655	0.3551
2.535	59.0609	20.9695	0.3550
2.540	59.0050	20.9895	0.3557
2.545	58.9690	20.9947	0.3560
2.550	58.9373	21.0344	0.3569
2.555	58.9224	21.0385	0.3571
2.560	58.8789	21.0898	0.3582
2.565	58.8313	21.0965	0.3586
2.570	58.8123	21.1043	0.3588
2.575	58.7955	21.1265	0.3593
2.580	58.7397	21.1444	0.3600
2.585	58.7344	21.2131	0.3612
2.590	58.6738	21.1671	0.3608
2.595	58.6506	21.2240	0.3619
2.600	58.5982	21.2298	0.3623
2.605	58.5259	21.2501	0.3631
2.610	58.5705	21.2502	0.3628
2.615	58.5092	21.2392	0.3630

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	58.4554	21.3130	0.3646
2.625	58.4217	21.3254	0.3650
2.630	58.4424	21.3453	0.3652
2.635	58.4070	21.3332	0.3653
2.640	58.3328	21.3929	0.3667
2.645	58.3361	21.3977	0.3668
2.650	58.2739	21.4486	0.3681
2.655	58.2650	21.4215	0.3677
2.660	58.2226	21.4702	0.3688
2.665	58.2435	21.4669	0.3686
2.670	58.1674	21.4471	0.3687
2.675	58.1287	21.5086	0.3700
2.680	58.1097	21.5190	0.3703
2.685	58.0464	21.5627	0.3715
2.690	58.0224	21.5823	0.3720
2.695	58.0079	21.6038	0.3724
2.700	57.9832	21.6414	0.3732
2.705	57.9624	21.6465	0.3735
2.710	57.9563	21.7266	0.3749
2.715	57.9091	21.6644	0.3741
2.720	57.8805	21.7383	0.3756
2.725	57.7793	21.7706	0.3768
2.730	57.7636	21.8202	0.3777
2.735	57.7632	21.8589	0.3784
2.740	57.7138	21.8005	0.3777
2.745	57.7222	21.9101	0.3796
2.750	57.6951	21.9309	0.3801
2.755	57.6557	21.9654	0.3810
2.760	57.6811	21.9677	0.3808
2.765	57.5919	21.9695	0.3815
2.770	57.6258	22.0165	0.3821
2.775	57.5452	22.0210	0.3827
2.780	57.5252	22.0270	0.3829
2.785	57.5214	22.0719	0.3837
2.790	57.4982	22.0445	0.3834
2.795	57.4734	22.1205	0.3849
2.800	57.4472	22.1346	0.3853
2.805	57.4397	22.1507	0.3856
2.810	57.3423	22.1839	0.3869
2.815	57.3538	22.2236	0.3875
2.820	57.3571	22.2053	0.3871
2.825	57.3029	22.2013	0.3874
2.830	57.2799	22.2504	0.3885
2.835	57.2713	22.2540	0.3886
2.840	57.2177	22.3326	0.3903
2.845	57.1985	22.3442	0.3906

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	57.1384	22.2785	0.3899
2.855	57.1596	22.3598	0.3912
2.860	57.0861	22.3559	0.3916
2.865	57.0341	22.4179	0.3931
2.870	57.0106	22.4252	0.3934
2.875	56.9975	22.4253	0.3934
2.880	56.9870	22.4955	0.3947
2.885	56.9663	22.4826	0.3947
2.890	56.9219	22.5893	0.3968
2.895	56.8835	22.5263	0.3960
2.900	56.8950	22.5478	0.3963
2.905	56.8812	22.6072	0.3974
2.910	56.8760	22.6218	0.3977
2.915	56.8184	22.6107	0.3979
2.920	56.8230	22.6275	0.3982
2.925	56.7874	22.5862	0.3977
2.930	56.7461	22.6336	0.3989
2.935	56.7132	22.6531	0.3994
2.940	56.7262	22.6689	0.3996
2.945	56.6778	22.7167	0.4008
2.950	56.6184	22.7211	0.4013
2.955	56.6753	22.6625	0.3999
2.960	56.5170	22.8356	0.4040
2.965	56.5680	22.7193	0.4016
2.970	56.4538	22.7406	0.4028
2.975	56.4477	22.7605	0.4032
2.980	56.4800	22.7835	0.4034
2.985	56.3845	22.8173	0.4047
2.990	56.3748	22.8932	0.4061
2.995	56.3229	22.8277	0.4053
3.000	56.2724	22.9292	0.4075
3.005	56.2837	22.8870	0.4066
3.010	56.1996	22.8556	0.4067
3.015	56.1892	22.8957	0.4075
3.020	56.1760	23.0403	0.4101
3.025	56.1311	23.0049	0.4098
3.030	56.1235	22.9745	0.4094
3.035	56.1843	23.0479	0.4102
3.040	56.0951	23.1018	0.4118
3.045	56.0587	23.0959	0.4120
3.050	55.9410	23.1250	0.4134
3.055	56.0636	23.1893	0.4136
3.060	56.0645	23.2369	0.4145
3.065	56.0945	23.1573	0.4128
3.070	56.0586	23.2903	0.4155
3.075	56.0619	23.2634	0.4150

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	56.0893	23.3096	0.4156
3.085	56.0021	23.2629	0.4154
3.090	56.0346	23.2681	0.4152
3.095	56.0225	23.2696	0.4154
3.100	56.0373	23.2426	0.4148
3.105	55.9379	23.2065	0.4149
3.110	55.9690	23.2539	0.4155
3.115	55.8997	23.2183	0.4154
3.120	55.9058	23.2693	0.4162
3.125	55.8395	23.2697	0.4167
3.130	55.8121	23.2690	0.4169
3.135	55.7588	23.3040	0.4179
3.140	55.7493	23.2939	0.4178
3.145	55.6565	23.3054	0.4187
3.150	55.6723	23.3206	0.4189
3.155	55.6135	23.3664	0.4202
3.160	55.6205	23.3396	0.4196
3.165	55.5300	23.4103	0.4216
3.170	55.5512	23.3556	0.4204
3.175	55.4505	23.3937	0.4219
3.180	55.4572	23.4540	0.4229
3.185	55.4025	23.5284	0.4247
3.190	55.4197	23.5204	0.4244
3.195	55.3737	23.5786	0.4258
3.200	55.3112	23.5954	0.4266
3.205	55.3875	23.6407	0.4268
3.210	55.3377	23.6462	0.4273
3.215	55.3219	23.6252	0.4270
3.220	55.3828	23.6222	0.4265
3.225	55.3219	23.7205	0.4288
3.230	55.2590	23.6134	0.4273
3.235	55.2671	23.7120	0.4290
3.240	55.2070	23.7372	0.4300
3.245	55.1808	23.6813	0.4292
3.250	55.1381	23.8396	0.4324
3.255	55.0478	23.8031	0.4324
3.260	55.1066	23.7624	0.4312
3.265	55.0781	23.8288	0.4326
3.270	54.9167	23.7441	0.4324
3.275	54.9320	23.8082	0.4334
3.280	54.9486	23.8149	0.4334
3.285	54.7993	23.9111	0.4363
3.290	54.8394	23.8015	0.4340
3.295	54.7550	23.8567	0.4357
3.300	54.7890	23.8431	0.4352
3.305	54.7424	23.9218	0.4370

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	54.6769	23.9388	0.4378
3.315	54.6116	23.9346	0.4383
3.320	54.6546	23.9844	0.4388
3.325	54.6488	23.9607	0.4384
3.330	54.5599	24.0522	0.4408
3.335	54.4469	24.0465	0.4417
3.340	54.4830	24.0574	0.4416
3.345	54.3671	24.0503	0.4424
3.350	54.3370	24.1264	0.4440
3.355	54.3618	24.0884	0.4431
3.360	54.3031	24.2007	0.4457
3.365	54.3003	24.2515	0.4466
3.370	54.2115	24.2782	0.4478
3.375	54.3056	24.2955	0.4474
3.380	54.2539	24.2796	0.4475
3.385	54.1772	24.3151	0.4488
3.390	54.1443	24.3317	0.4494
3.395	54.1683	24.3366	0.4493
3.400	54.0826	24.4285	0.4517
3.405	54.0429	24.3948	0.4514
3.410	54.0682	24.4393	0.4520
3.415	53.9900	24.4847	0.4535
3.420	53.9895	24.4447	0.4528
3.425	54.0242	24.4144	0.4519
3.430	54.0184	24.4382	0.4524
3.435	54.0003	24.5111	0.4539
3.440	53.9350	24.4564	0.4534
3.445	53.8874	24.4742	0.4542
3.450	53.7952	24.5101	0.4556
3.455	53.8757	24.5484	0.4556
3.460	53.7879	24.5657	0.4567
3.465	53.7281	24.5716	0.4573
3.470	53.7332	24.5317	0.4565
3.475	53.6922	24.4758	0.4559
3.480	53.6063	24.4884	0.4568
3.485	53.5557	24.3477	0.4546
3.490	53.5550	24.5844	0.4591
3.495	53.4970	24.6048	0.4599
3.500	53.4319	24.5661	0.4598
3.505	53.5143	24.5925	0.4596
3.510	53.3621	24.5418	0.4599
3.515	53.3133	24.6660	0.4627
3.520	53.2325	24.6935	0.4639
3.525	53.1326	24.8966	0.4686
3.530	53.2004	24.9540	0.4691
3.535	53.0918	24.8950	0.4689

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	53.0873	25.0406	0.4717
3.545	53.2421	25.0805	0.4711
3.550	53.1944	25.1770	0.4733
3.555	53.2450	25.1213	0.4718
3.560	53.3267	25.1625	0.4719
3.565	53.3729	25.1081	0.4704
3.570	53.3095	25.2152	0.4730
3.575	53.4611	25.3482	0.4741
3.580	53.5666	25.1538	0.4696
3.585	53.5309	24.8605	0.4644
3.590	53.4337	24.6880	0.4620
3.595	53.3432	24.6102	0.4614
3.600	53.2705	24.7158	0.4640
3.605	53.1176	24.5718	0.4626
3.610	53.0297	24.6183	0.4642
3.615	52.8654	24.7073	0.4674
3.620	52.7614	24.7413	0.4689
3.625	52.7688	24.6096	0.4664
3.630	52.4465	24.8662	0.4741
3.635	52.5481	24.9857	0.4755
3.640	52.6155	25.0520	0.4761
3.645	52.5842	25.1439	0.4782
3.650	52.5358	25.2270	0.4802
3.655	52.5100	25.2015	0.4799
3.660	52.5607	25.3281	0.4819
3.665	52.5328	25.2825	0.4813
3.670	52.5155	25.3668	0.4830
3.675	52.5134	25.3464	0.4827
3.680	52.5632	25.3728	0.4827
3.685	52.6208	25.4602	0.4838
3.690	52.6190	25.4413	0.4835
3.695	52.6678	25.4503	0.4832
3.700	52.6939	25.4313	0.4826
3.705	52.7240	25.3328	0.4805
3.710	52.6669	25.3599	0.4815
3.715	52.7168	25.3053	0.4800
3.720	52.7107	25.1776	0.4777
3.725	52.6661	25.2155	0.4788
3.730	52.6046	25.1100	0.4773
3.735	52.4243	25.0646	0.4781
3.740	52.3753	25.0353	0.4780
3.745	52.2789	25.0528	0.4792
3.750	52.1228	25.1283	0.4821
3.755	52.0211	25.2373	0.4851
3.760	51.9511	25.4006	0.4889
3.765	51.9259	25.6002	0.4930

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	52.1013	25.6978	0.4932
3.775	52.1724	25.7131	0.4928
3.780	52.1154	26.0734	0.5003
3.785	52.2916	25.7412	0.4923
3.790	52.2961	25.5182	0.4880
3.795	52.2433	25.5236	0.4886
3.800	52.2666	25.6115	0.4900
3.805	52.2557	25.4625	0.4873
3.810	52.1316	25.4894	0.4889
3.815	51.9923	25.4181	0.4889
3.820	52.0751	25.4748	0.4892
3.825	51.9506	25.4155	0.4892
3.830	51.9674	25.4955	0.4906
3.835	50.9578	25.1914	0.4944
3.840	51.8267	25.5326	0.4927
3.845	51.7930	25.5557	0.4934
3.850	51.7760	25.5477	0.4934
3.855	51.7423	25.6199	0.4951
3.860	51.6661	25.5953	0.4954
3.865	51.5887	25.6609	0.4974
3.870	51.6628	25.6193	0.4959
3.875	51.5844	25.6540	0.4973
3.880	51.6084	25.6399	0.4968
3.885	51.5280	25.7064	0.4989
3.890	51.5818	25.7164	0.4986
3.895	51.5134	25.6251	0.4974
3.900	51.4723	25.6871	0.4990
3.905	51.4160	25.8003	0.5018
3.910	51.4275	25.7420	0.5006
3.915	51.3795	25.7557	0.5013
3.920	51.3853	25.8157	0.5024
3.925	51.3049	25.7847	0.5026
3.930	51.2910	25.7877	0.5028
3.935	51.2685	25.7602	0.5025
3.940	51.2460	25.8220	0.5039
3.945	51.1915	25.7731	0.5035
3.950	51.1379	25.8393	0.5053
3.955	51.1223	25.8747	0.5061
3.960	51.0946	25.8465	0.5059
3.965	51.1077	25.8469	0.5057
3.970	51.0560	25.9262	0.5078
3.975	51.0021	25.9050	0.5079
3.980	50.9793	25.9099	0.5082
3.985	50.9864	25.9491	0.5089
3.990	50.9532	25.9547	0.5094
3.995	50.8745	25.9454	0.5100

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	50.8575	25.9805	0.5108
4.005	50.8416	25.9764	0.5109
4.010	50.7706	26.0018	0.5121
4.015	50.7887	26.0606	0.5131
4.020	50.7327	26.0267	0.5130
4.025	50.7396	25.9797	0.5120
4.030	50.6476	26.0550	0.5144
4.035	50.6361	26.0856	0.5152
4.040	50.6463	26.0596	0.5145
4.045	50.6080	26.0453	0.5146
4.050	50.5513	26.0471	0.5153
4.055	50.4845	26.0672	0.5163
4.060	50.4526	26.1167	0.5176
4.065	50.3985	26.1195	0.5183
4.070	50.4465	26.1620	0.5186
4.075	50.3682	26.1475	0.5191
4.080	50.3611	26.1995	0.5202
4.085	50.3184	26.1918	0.5205
4.090	50.3098	26.1940	0.5207
4.095	50.2667	26.2653	0.5225
4.100	50.2858	26.2430	0.5219
4.105	50.1163	26.1661	0.5221
4.110	50.2227	26.2394	0.5225
4.115	50.1374	26.2541	0.5236
4.120	50.0971	26.2846	0.5247
4.125	50.1400	26.2074	0.5227
4.130	50.1629	26.2864	0.5240
4.135	50.0196	26.2622	0.5250
4.140	49.9707	26.1963	0.5242
4.145	50.0590	26.1659	0.5227
4.150	49.9079	26.2984	0.5269
4.155	49.8914	26.3135	0.5274
4.160	49.9326	26.2765	0.5262
4.165	49.8362	26.2629	0.5270
4.170	49.8116	26.2762	0.5275
4.175	49.8558	26.2025	0.5256
4.180	49.8607	26.2624	0.5267
4.185	49.7902	26.2275	0.5268
4.190	49.7246	26.2513	0.5279
4.195	49.7240	26.2991	0.5289
4.200	49.7722	26.3120	0.5286
4.205	49.7940	26.3182	0.5285
4.210	49.6648	26.4202	0.5320
4.215	49.7307	26.3897	0.5307
4.220	49.6897	26.4236	0.5318
4.225	49.7931	26.3839	0.5299

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	49.6824	26.4238	0.5319
4.235	49.6173	26.4699	0.5335
4.240	49.6671	26.5614	0.5348
4.245	49.6756	26.4851	0.5332
4.250	49.5957	26.4992	0.5343
4.255	49.6528	26.5355	0.5344
4.260	49.5674	26.5830	0.5363
4.265	49.6194	26.5436	0.5349
4.270	49.5485	26.5941	0.5367
4.275	49.5265	26.6090	0.5373
4.280	49.5202	26.6385	0.5379
4.285	49.5061	26.6717	0.5388
4.290	49.4527	26.6821	0.5395
4.295	49.4570	26.6984	0.5398
4.300	49.4499	26.7183	0.5403
4.305	49.4235	26.7300	0.5408
4.310	49.4072	26.7190	0.5408
4.315	49.3865	26.7392	0.5414
4.320	49.3840	26.7274	0.5412
4.325	49.3678	26.7598	0.5420
4.330	49.3533	26.7602	0.5422
4.335	49.3331	26.7704	0.5426
4.340	49.3094	26.7445	0.5424
4.345	49.2593	26.7605	0.5433
4.350	49.2564	26.7544	0.5432
4.355	49.2283	26.8013	0.5444
4.360	49.1626	26.7872	0.5449
4.365	49.1070	26.8242	0.5462
4.370	49.1337	26.7976	0.5454
4.375	49.0865	26.8899	0.5478
4.380	49.0637	26.8679	0.5476
4.385	48.9462	26.8936	0.5495
4.390	48.9589	26.8569	0.5486
4.395	48.9082	26.9196	0.5504
4.400	48.8916	26.9526	0.5513
4.405	48.8189	26.9781	0.5526
4.410	48.8547	26.9908	0.5525
4.415	48.8098	26.9864	0.5529
4.420	48.7968	27.0643	0.5546
4.425	48.8191	27.0102	0.5533
4.430	48.7218	27.0259	0.5547
4.435	48.6508	27.0326	0.5556
4.440	48.6984	27.0510	0.5555
4.445	48.5726	27.0962	0.5579
4.450	48.5933	27.0100	0.5558
4.455	48.4914	27.1087	0.5590

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	48.4931	26.9792	0.5564
4.465	48.4428	26.9164	0.5556
4.470	48.4417	26.9186	0.5557
4.475	48.4492	26.8788	0.5548
4.480	48.4889	26.8653	0.5541
4.485	48.3027	26.8573	0.5560
4.490	48.3611	26.9141	0.5565
4.495	48.3583	26.9125	0.5565
4.500	48.3226	26.9343	0.5574
4.505	48.2396	26.9752	0.5592
4.510	48.1643	27.0274	0.5611
4.515	48.2354	27.0757	0.5613
4.520	48.2026	27.0959	0.5621
4.525	48.2075	27.0729	0.5616
4.530	48.0811	27.1017	0.5637
4.535	48.0996	27.1308	0.5641
4.540	48.0850	27.0928	0.5634
4.545	48.0183	27.1240	0.5649
4.550	48.0015	27.0697	0.5639
4.555	47.9896	27.1389	0.5655
4.560	47.9581	27.1343	0.5658
4.565	47.9665	27.2002	0.5671
4.570	47.9223	27.1664	0.5669
4.575	47.9098	27.1246	0.5662
4.580	47.9181	27.1738	0.5671
4.585	47.8810	27.1722	0.5675
4.590	47.8015	27.1679	0.5683
4.595	47.8151	27.2342	0.5696
4.600	47.7763	27.2067	0.5695
4.605	47.7362	27.2329	0.5705
4.610	47.7294	27.2196	0.5703
4.615	47.6927	27.1888	0.5701
4.620	47.6443	27.1899	0.5707
4.625	47.6509	27.2174	0.5712
4.630	47.6033	27.2003	0.5714
4.635	47.5325	27.1937	0.5721
4.640	47.5067	27.2274	0.5731
4.645	47.4935	27.2041	0.5728
4.650	47.4583	27.2247	0.5737
4.655	47.3899	27.1803	0.5735
4.660	47.4763	27.2021	0.5730
4.665	47.4260	27.2074	0.5737
4.670	47.2813	27.2853	0.5771
4.675	47.2342	27.1895	0.5756
4.680	47.2460	27.2153	0.5760
4.685	47.2338	27.2142	0.5762

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	47.2658	27.3033	0.5777
4.695	47.1962	27.1756	0.5758
4.700	47.2008	27.2791	0.5779
4.705	47.0332	27.1724	0.5777
4.710	46.9876	27.3944	0.5830
4.715	47.0641	27.3502	0.5811
4.720	47.0180	27.3019	0.5807
4.725	47.0737	27.4117	0.5823
4.730	47.0482	27.3424	0.5812
4.735	47.0675	27.4003	0.5821
4.740	47.0480	27.3541	0.5814
4.745	46.9822	27.4409	0.5841
4.750	47.0336	27.3300	0.5811
4.755	47.0452	27.3820	0.5820
4.760	46.9678	27.3693	0.5827
4.765	47.0287	27.3327	0.5812
4.770	47.0257	27.3917	0.5825
4.775	46.8021	27.3755	0.5849
4.780	46.8355	27.3776	0.5845
4.785	46.9845	27.5814	0.5870
4.790	46.7890	27.3476	0.5845
4.795	46.7287	27.4475	0.5874
4.800	46.7307	27.4726	0.5879
4.805	46.7185	27.3487	0.5854
4.810	46.7186	27.4948	0.5885
4.815	46.6994	27.3930	0.5866
4.820	46.6374	27.6083	0.5920
4.825	46.6665	27.5449	0.5902
4.830	46.6141	27.6354	0.5929
4.835	46.6286	27.6547	0.5931
4.840	46.6288	27.6134	0.5922
4.845	46.6176	27.5481	0.5909
4.850	46.5723	27.6673	0.5941
4.855	46.6717	27.6535	0.5925
4.860	46.5725	27.6499	0.5937
4.865	46.6309	27.5881	0.5916
4.870	46.5521	27.6401	0.5937
4.875	46.5416	27.6656	0.5944
4.880	46.5538	27.7079	0.5952
4.885	46.4716	27.6702	0.5954
4.890	46.5371	27.6697	0.5946
4.895	46.4020	27.6902	0.5967
4.900	46.4434	27.6850	0.5961
4.905	46.4053	27.7007	0.5969
4.910	46.3605	27.6975	0.5974
4.915	46.3502	27.7082	0.5978

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	46.3340	27.6985	0.5978
4.925	46.3228	27.6783	0.5975
4.930	46.2634	27.6907	0.5985
4.935	46.2680	27.6797	0.5982
4.940	46.2587	27.6827	0.5984
4.945	46.2336	27.7055	0.5993
4.950	46.2468	27.7117	0.5992
4.955	46.2501	27.7275	0.5995
4.960	46.1911	27.7180	0.6001
4.965	46.1971	27.7384	0.6004
4.970	46.2006	27.7787	0.6013
4.975	46.2097	27.7584	0.6007
4.980	46.1432	27.8326	0.6032
4.985	46.1089	27.8240	0.6034
4.990	46.1170	27.8868	0.6047
4.995	46.0307	27.9589	0.6074
5.000	46.0055	27.9567	0.6077
5.005	45.9603	28.0056	0.6093
5.010	45.8983	28.0269	0.6106
5.015	45.8759	28.1038	0.6126
5.020	45.7773	28.0904	0.6136
5.025	45.6632	28.1610	0.6167
5.030	45.5561	28.1611	0.6182
5.035	45.5373	28.1727	0.6187
5.040	45.4195	28.1490	0.6198
5.045	45.3639	28.0726	0.6188
5.050	45.2784	27.9960	0.6183
5.055	45.3173	27.9687	0.6172
5.060	45.2539	27.9629	0.6179
5.065	45.2416	27.8728	0.6161
5.070	45.2613	27.9189	0.6168
5.075	45.2352	27.8525	0.6157
5.080	45.1828	27.8597	0.6166
5.085	45.1690	27.8202	0.6159
5.090	45.2211	27.8216	0.6152
5.095	45.2785	27.8606	0.6153
5.100	45.2795	27.8442	0.6149
5.105	45.3094	27.9176	0.6162
5.110	45.2136	27.8701	0.6164
5.115	45.1898	27.8790	0.6169
5.120	45.1854	27.8856	0.6171
5.125	45.1635	27.9041	0.6178
5.130	45.1434	27.9126	0.6183
5.135	45.0638	27.9549	0.6203
5.140	45.1034	27.9830	0.6204
5.145	45.0024	27.9493	0.6211

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	45.0278	28.0178	0.6222
5.155	44.9588	27.9734	0.6222
5.160	44.9265	27.9713	0.6226
5.165	44.8948	27.9894	0.6234
5.170	44.8623	27.9733	0.6235
5.175	44.7877	27.9491	0.6240
5.180	44.7993	27.9523	0.6239
5.185	44.7552	27.9903	0.6254
5.190	44.7662	27.9245	0.6238
5.195	44.7496	27.9307	0.6242
5.200	44.7559	27.9286	0.6240
5.205	44.7340	27.9390	0.6246
5.210	44.7244	27.9275	0.6244
5.215	44.7365	27.9484	0.6247
5.220	44.6822	27.9500	0.6255
5.225	44.6184	27.9307	0.6260
5.230	44.6341	27.9390	0.6260
5.235	44.6197	27.9746	0.6270
5.240	44.5660	27.9783	0.6278
5.245	44.5876	27.9776	0.6275
5.250	44.5546	28.0360	0.6293
5.255	44.5032	28.0607	0.6305
5.260	44.5469	27.9536	0.6275
5.265	44.5991	28.0389	0.6287
5.270	44.4743	28.0237	0.6301
5.275	44.4323	27.9856	0.6298
5.280	44.4644	28.0357	0.6305
5.285	44.4507	27.9735	0.6293
5.290	44.3724	27.9793	0.6306
5.295	44.3713	27.8850	0.6284
5.300	44.3566	27.9800	0.6308
5.305	44.3854	27.9578	0.6299
5.310	44.2298	28.0320	0.6338
5.315	44.2874	28.0066	0.6324
5.320	44.2004	27.9778	0.6330
5.325	44.2349	27.9957	0.6329
5.330	44.1984	28.0182	0.6339
5.335	44.1278	28.0779	0.6363
5.340	44.1445	28.1332	0.6373
5.345	44.1315	28.1379	0.6376
5.350	44.1446	28.0434	0.6353
5.355	44.0686	28.1171	0.6380
5.360	43.9851	28.0346	0.6374
5.365	44.1236	28.0449	0.6356
5.370	43.9324	28.0651	0.6388
5.375	44.0265	28.0961	0.6382

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	44.0129	28.0971	0.6384
5.385	43.9477	28.0960	0.6393
5.390	43.9647	28.1914	0.6412
5.395	44.0018	28.1490	0.6397
5.400	43.9064	28.1683	0.6416
5.405	43.8824	28.1564	0.6416
5.410	43.8815	28.1943	0.6425
5.415	43.8643	28.0993	0.6406
5.420	43.8316	28.2120	0.6436
5.425	43.8399	28.2370	0.6441
5.430	43.7972	28.1792	0.6434
5.435	43.7922	28.1914	0.6438
5.440	43.7282	28.2856	0.6468
5.445	43.7879	28.2205	0.6445
5.450	43.6891	28.2747	0.6472
5.455	43.7496	28.2495	0.6457
5.460	43.6810	28.2576	0.6469
5.465	43.7591	28.3140	0.6470
5.470	43.6231	28.3471	0.6498
5.475	43.6285	28.2925	0.6485
5.480	43.6219	28.3780	0.6505
5.485	43.6045	28.3345	0.6498
5.490	43.5814	28.4272	0.6523
5.495	43.6157	28.3215	0.6493
5.500	43.5907	28.3798	0.6511
5.505	43.5977	28.3783	0.6509
5.510	43.5765	28.3683	0.6510
5.515	43.5562	28.3371	0.6506
5.520	43.5005	28.3682	0.6521
5.525	43.5245	28.3462	0.6513
5.530	43.5094	28.3636	0.6519
5.535	43.4929	28.3310	0.6514
5.540	43.4749	28.3803	0.6528
5.545	43.4176	28.3668	0.6533
5.550	43.4290	28.3673	0.6532
5.555	43.4212	28.3610	0.6532
5.560	43.4056	28.3841	0.6539
5.565	43.4171	28.3619	0.6532
5.570	43.3997	28.3746	0.6538
5.575	43.3945	28.3664	0.6537
5.580	43.4137	28.3947	0.6540
5.585	43.3777	28.3822	0.6543
5.590	43.3692	28.3537	0.6538
5.595	43.3466	28.3600	0.6543
5.600	43.3507	28.3510	0.6540
5.605	43.3393	28.2973	0.6529

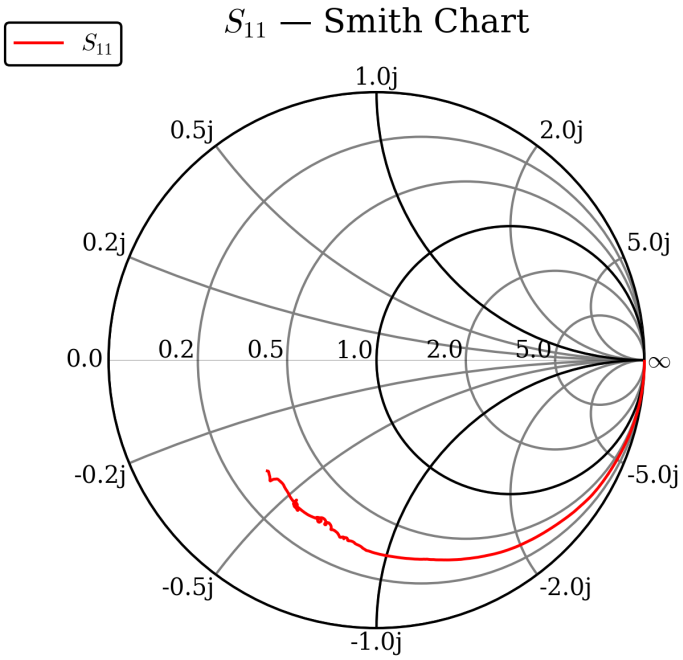
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	43.2657	28.2893	0.6539
5.615	43.2274	28.2606	0.6538
5.620	43.1977	28.2591	0.6542
5.625	43.1965	28.3157	0.6555
5.630	43.1172	28.2737	0.6557
5.635	43.1126	28.2394	0.6550
5.640	43.0572	28.2208	0.6554
5.645	42.9940	28.2475	0.6570
5.650	43.0427	28.2639	0.6566
5.655	42.9896	28.2731	0.6577
5.660	42.9586	28.3152	0.6591
5.665	43.0132	28.2353	0.6564
5.670	42.8694	28.3574	0.6615
5.675	42.8601	28.2386	0.6589
5.680	42.8146	28.2705	0.6603
5.685	42.8417	28.2005	0.6582
5.690	42.7796	28.2166	0.6596
5.695	42.7982	28.3092	0.6615
5.700	42.7000	28.2323	0.6612
5.705	42.6061	28.2816	0.6638
5.710	42.6786	28.2190	0.6612
5.715	42.5860	28.2267	0.6628
5.720	42.5722	28.2724	0.6641
5.725	42.5613	28.3250	0.6655
5.730	42.5146	28.2828	0.6652
5.735	42.5172	28.2957	0.6655
5.740	42.5028	28.3182	0.6663
5.745	42.4997	28.3158	0.6663
5.750	42.4137	28.3315	0.6680
5.755	42.4079	28.3122	0.6676
5.760	42.3263	28.3230	0.6692
5.765	42.3757	28.2596	0.6669
5.770	42.2816	28.2683	0.6686
5.775	42.3050	28.2857	0.6686
5.780	42.2853	28.3665	0.6708
5.785	42.2095	28.3286	0.6711
5.790	42.1793	28.3887	0.6730
5.795	42.2028	28.4021	0.6730
5.800	42.1547	28.3760	0.6731
5.805	42.1154	28.3635	0.6735
5.810	42.1437	28.4267	0.6745
5.815	42.1443	28.4250	0.6745
5.820	42.0826	28.4099	0.6751
5.825	42.0686	28.4021	0.6751
5.830	42.0190	28.4378	0.6768
5.835	42.0046	28.4118	0.6764

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	42.0129	28.4217	0.6765
5.845	41.9677	28.4571	0.6781
5.850	41.9414	28.4131	0.6774
5.855	41.9328	28.4220	0.6778
5.860	41.8856	28.4399	0.6790
5.865	41.8950	28.4552	0.6792
5.870	41.8506	28.4273	0.6793
5.875	41.8345	28.4659	0.6804
5.880	41.8212	28.4619	0.6806
5.885	41.7981	28.3910	0.6792
5.890	41.7680	28.5039	0.6824
5.895	41.7386	28.4760	0.6822
5.900	41.6970	28.4257	0.6817
5.905	41.7199	28.5027	0.6832
5.910	41.6209	28.4343	0.6832
5.915	41.6401	28.4574	0.6834
5.920	41.6004	28.4323	0.6835
5.925	41.5638	28.4482	0.6844
5.930	41.5331	28.4972	0.6861
5.935	41.4874	28.4570	0.6859
5.940	41.4812	28.4951	0.6869
5.945	41.5440	28.4378	0.6845
5.950	41.4856	28.5109	0.6872
5.955	41.4723	28.5218	0.6877
5.960	41.4257	28.5207	0.6885
5.965	41.4008	28.5136	0.6887
5.970	41.4192	28.5536	0.6894
5.975	41.3624	28.3819	0.6862
5.980	41.5404	28.4338	0.6845
5.985	41.3578	28.5321	0.6899
5.990	41.2770	28.4732	0.6898
5.995	41.2661	28.4529	0.6895
6.000	41.2898	28.4080	0.6880

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0044	-0.00
0.010	0.9999	-0.0088	-0.00
0.015	0.9998	-0.0131	-0.00
0.020	0.9997	-0.0176	-0.00
0.025	0.9995	-0.0220	-0.00
0.030	0.9995	-0.0266	-0.00
0.035	0.9994	-0.0310	-0.00
0.040	0.9990	-0.0354	-0.00
0.045	0.9988	-0.0398	-0.00
0.050	0.9985	-0.0443	-0.00
0.055	0.9984	-0.0486	-0.00
0.060	0.9980	-0.0531	-0.01
0.065	0.9977	-0.0576	-0.01
0.070	0.9972	-0.0620	-0.01
0.075	0.9970	-0.0662	-0.01
0.080	0.9964	-0.0706	-0.01
0.085	0.9960	-0.0751	-0.01
0.090	0.9954	-0.0795	-0.01
0.095	0.9949	-0.0840	-0.01
0.100	0.9944	-0.0882	-0.01
0.105	0.9938	-0.0927	-0.02
0.110	0.9934	-0.0971	-0.02
0.115	0.9927	-0.1015	-0.02

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9922	-0.1056	-0.02
0.125	0.9914	-0.1102	-0.02
0.130	0.9905	-0.1144	-0.03
0.135	0.9901	-0.1188	-0.02
0.140	0.9892	-0.1233	-0.03
0.145	0.9885	-0.1276	-0.03
0.150	0.9876	-0.1319	-0.03
0.155	0.9870	-0.1362	-0.03
0.160	0.9861	-0.1404	-0.03
0.165	0.9854	-0.1447	-0.04
0.170	0.9845	-0.1490	-0.04
0.175	0.9835	-0.1533	-0.04
0.180	0.9824	-0.1575	-0.04
0.185	0.9816	-0.1619	-0.05
0.190	0.9805	-0.1661	-0.05
0.195	0.9795	-0.1704	-0.05
0.200	0.9785	-0.1746	-0.05
0.205	0.9773	-0.1789	-0.06
0.210	0.9762	-0.1831	-0.06
0.215	0.9752	-0.1873	-0.06
0.220	0.9739	-0.1915	-0.06
0.225	0.9729	-0.1958	-0.07
0.230	0.9716	-0.1999	-0.07
0.235	0.9703	-0.2042	-0.07
0.240	0.9691	-0.2084	-0.08
0.245	0.9679	-0.2125	-0.08
0.250	0.9665	-0.2167	-0.08
0.255	0.9653	-0.2208	-0.09
0.260	0.9638	-0.2251	-0.09
0.265	0.9624	-0.2291	-0.09
0.270	0.9610	-0.2333	-0.10
0.275	0.9597	-0.2374	-0.10
0.280	0.9582	-0.2414	-0.10
0.285	0.9567	-0.2455	-0.11
0.290	0.9554	-0.2496	-0.11
0.295	0.9538	-0.2536	-0.11
0.300	0.9521	-0.2576	-0.12
0.305	0.9507	-0.2617	-0.12
0.310	0.9491	-0.2657	-0.13
0.315	0.9475	-0.2697	-0.13
0.320	0.9460	-0.2736	-0.13
0.325	0.9442	-0.2776	-0.14
0.330	0.9426	-0.2816	-0.14
0.335	0.9410	-0.2855	-0.15
0.340	0.9392	-0.2894	-0.15
0.345	0.9376	-0.2932	-0.15

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9358	-0.2971	-0.16
0.355	0.9341	-0.3010	-0.16
0.360	0.9323	-0.3048	-0.17
0.365	0.9305	-0.3086	-0.17
0.370	0.9286	-0.3124	-0.18
0.375	0.9266	-0.3162	-0.18
0.380	0.9249	-0.3201	-0.19
0.385	0.9231	-0.3239	-0.19
0.390	0.9212	-0.3277	-0.20
0.395	0.9194	-0.3312	-0.20
0.400	0.9176	-0.3349	-0.20
0.405	0.9157	-0.3386	-0.21
0.410	0.9137	-0.3423	-0.21
0.415	0.9118	-0.3459	-0.22
0.420	0.9100	-0.3496	-0.22
0.425	0.9080	-0.3531	-0.23
0.430	0.9060	-0.3566	-0.23
0.435	0.9041	-0.3602	-0.24
0.440	0.9022	-0.3637	-0.24
0.445	0.9003	-0.3672	-0.24
0.450	0.8982	-0.3708	-0.25
0.455	0.8962	-0.3743	-0.25
0.460	0.8941	-0.3777	-0.26
0.465	0.8922	-0.3812	-0.26
0.470	0.8900	-0.3845	-0.27
0.475	0.8880	-0.3880	-0.27
0.480	0.8859	-0.3913	-0.28
0.485	0.8840	-0.3947	-0.28
0.490	0.8819	-0.3979	-0.29
0.495	0.8798	-0.4014	-0.29
0.500	0.8777	-0.4046	-0.30
0.505	0.8757	-0.4079	-0.30
0.510	0.8735	-0.4111	-0.31
0.515	0.8715	-0.4145	-0.31
0.520	0.8694	-0.4177	-0.31
0.525	0.8675	-0.4208	-0.32
0.530	0.8654	-0.4240	-0.32
0.535	0.8632	-0.4272	-0.33
0.540	0.8612	-0.4303	-0.33
0.545	0.8590	-0.4335	-0.33
0.550	0.8568	-0.4366	-0.34
0.555	0.8545	-0.4399	-0.34
0.560	0.8524	-0.4429	-0.35
0.565	0.8502	-0.4460	-0.35
0.570	0.8479	-0.4490	-0.36
0.575	0.8457	-0.4522	-0.36

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8434	-0.4551	-0.37
0.585	0.8413	-0.4581	-0.37
0.590	0.8390	-0.4611	-0.38
0.595	0.8369	-0.4640	-0.38
0.600	0.8346	-0.4670	-0.39
0.605	0.8324	-0.4698	-0.39
0.610	0.8301	-0.4729	-0.40
0.615	0.8277	-0.4757	-0.40
0.620	0.8255	-0.4787	-0.41
0.625	0.8231	-0.4815	-0.41
0.630	0.8208	-0.4844	-0.42
0.635	0.8185	-0.4871	-0.42
0.640	0.8162	-0.4900	-0.43
0.645	0.8138	-0.4927	-0.43
0.650	0.8115	-0.4955	-0.44
0.655	0.8090	-0.4980	-0.45
0.660	0.8067	-0.5008	-0.45
0.665	0.8042	-0.5035	-0.46
0.670	0.8017	-0.5062	-0.46
0.675	0.7993	-0.5088	-0.47
0.680	0.7970	-0.5113	-0.47
0.685	0.7944	-0.5139	-0.48
0.690	0.7918	-0.5163	-0.49
0.695	0.7893	-0.5188	-0.50
0.700	0.7867	-0.5212	-0.50
0.705	0.7840	-0.5236	-0.51
0.710	0.7817	-0.5262	-0.52
0.715	0.7791	-0.5285	-0.52
0.720	0.7766	-0.5308	-0.53
0.725	0.7740	-0.5331	-0.54
0.730	0.7714	-0.5355	-0.55
0.735	0.7690	-0.5378	-0.55
0.740	0.7663	-0.5398	-0.56
0.745	0.7639	-0.5421	-0.57
0.750	0.7615	-0.5443	-0.57
0.755	0.7590	-0.5465	-0.58
0.760	0.7565	-0.5486	-0.59
0.765	0.7540	-0.5508	-0.59
0.770	0.7515	-0.5530	-0.60
0.775	0.7491	-0.5554	-0.61
0.780	0.7466	-0.5577	-0.61
0.785	0.7439	-0.5600	-0.62
0.790	0.7415	-0.5623	-0.63
0.795	0.7389	-0.5645	-0.63
0.800	0.7363	-0.5667	-0.64
0.805	0.7334	-0.5690	-0.65

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7308	-0.5710	-0.65
0.815	0.7280	-0.5732	-0.66
0.820	0.7254	-0.5754	-0.67
0.825	0.7226	-0.5776	-0.68
0.830	0.7199	-0.5795	-0.69
0.835	0.7171	-0.5816	-0.69
0.840	0.7145	-0.5837	-0.70
0.845	0.7118	-0.5857	-0.71
0.850	0.7090	-0.5877	-0.72
0.855	0.7063	-0.5897	-0.72
0.860	0.7036	-0.5917	-0.73
0.865	0.7008	-0.5936	-0.74
0.870	0.6981	-0.5957	-0.75
0.875	0.6953	-0.5976	-0.75
0.880	0.6926	-0.5995	-0.76
0.885	0.6899	-0.6014	-0.77
0.890	0.6870	-0.6034	-0.78
0.895	0.6843	-0.6054	-0.78
0.900	0.6815	-0.6073	-0.79
0.905	0.6788	-0.6093	-0.80
0.910	0.6760	-0.6112	-0.81
0.915	0.6732	-0.6131	-0.81
0.920	0.6704	-0.6150	-0.82
0.925	0.6676	-0.6169	-0.83
0.930	0.6648	-0.6188	-0.84
0.935	0.6619	-0.6207	-0.84
0.940	0.6590	-0.6225	-0.85
0.945	0.6562	-0.6243	-0.86
0.950	0.6533	-0.6262	-0.87
0.955	0.6505	-0.6280	-0.87
0.960	0.6477	-0.6300	-0.88
0.965	0.6448	-0.6317	-0.89
0.970	0.6420	-0.6336	-0.90
0.975	0.6392	-0.6354	-0.90
0.980	0.6364	-0.6370	-0.91
0.985	0.6336	-0.6389	-0.92
0.990	0.6307	-0.6405	-0.93
0.995	0.6279	-0.6423	-0.93
1.000	0.6250	-0.6440	-0.94
1.005	0.6221	-0.6457	-0.95
1.010	0.6193	-0.6473	-0.96
1.015	0.6164	-0.6490	-0.96
1.020	0.6134	-0.6507	-0.97
1.025	0.6106	-0.6524	-0.98
1.030	0.6076	-0.6541	-0.99
1.035	0.6049	-0.6556	-0.99

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.6020	-0.6573	-1.00
1.045	0.5993	-0.6588	-1.01
1.050	0.5963	-0.6603	-1.01
1.055	0.5937	-0.6618	-1.02
1.060	0.5908	-0.6633	-1.03
1.065	0.5880	-0.6648	-1.04
1.070	0.5852	-0.6665	-1.04
1.075	0.5824	-0.6679	-1.05
1.080	0.5796	-0.6694	-1.06
1.085	0.5769	-0.6708	-1.06
1.090	0.5740	-0.6721	-1.07
1.095	0.5712	-0.6735	-1.08
1.100	0.5685	-0.6748	-1.09
1.105	0.5658	-0.6762	-1.09
1.110	0.5631	-0.6775	-1.10
1.115	0.5605	-0.6789	-1.11
1.120	0.5578	-0.6801	-1.11
1.125	0.5550	-0.6814	-1.12
1.130	0.5523	-0.6826	-1.13
1.135	0.5497	-0.6839	-1.14
1.140	0.5473	-0.6851	-1.14
1.145	0.5447	-0.6863	-1.15
1.150	0.5420	-0.6875	-1.16
1.155	0.5394	-0.6886	-1.16
1.160	0.5368	-0.6898	-1.17
1.165	0.5343	-0.6911	-1.17
1.170	0.5318	-0.6922	-1.18
1.175	0.5294	-0.6932	-1.19
1.180	0.5268	-0.6943	-1.19
1.185	0.5244	-0.6954	-1.20
1.190	0.5222	-0.6964	-1.21
1.195	0.5197	-0.6973	-1.21
1.200	0.5173	-0.6983	-1.22
1.205	0.5149	-0.6993	-1.23
1.210	0.5126	-0.7001	-1.23
1.215	0.5101	-0.7010	-1.24
1.220	0.5076	-0.7019	-1.25
1.225	0.5054	-0.7029	-1.25
1.230	0.5028	-0.7036	-1.26
1.235	0.5005	-0.7045	-1.27
1.240	0.4981	-0.7055	-1.27
1.245	0.4958	-0.7063	-1.28
1.250	0.4935	-0.7071	-1.29
1.255	0.4911	-0.7080	-1.29
1.260	0.4890	-0.7087	-1.30
1.265	0.4867	-0.7094	-1.31

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.4845	-0.7103	-1.31
1.275	0.4821	-0.7109	-1.32
1.280	0.4799	-0.7118	-1.33
1.285	0.4775	-0.7126	-1.33
1.290	0.4751	-0.7133	-1.34
1.295	0.4727	-0.7140	-1.35
1.300	0.4705	-0.7147	-1.35
1.305	0.4682	-0.7155	-1.36
1.310	0.4661	-0.7160	-1.37
1.315	0.4638	-0.7168	-1.37
1.320	0.4617	-0.7175	-1.38
1.325	0.4595	-0.7179	-1.39
1.330	0.4572	-0.7186	-1.39
1.335	0.4551	-0.7192	-1.40
1.340	0.4528	-0.7199	-1.41
1.345	0.4506	-0.7204	-1.41
1.350	0.4483	-0.7209	-1.42
1.355	0.4461	-0.7216	-1.43
1.360	0.4438	-0.7222	-1.44
1.365	0.4415	-0.7228	-1.44
1.370	0.4393	-0.7233	-1.45
1.375	0.4370	-0.7238	-1.46
1.380	0.4346	-0.7242	-1.47
1.385	0.4322	-0.7248	-1.47
1.390	0.4299	-0.7255	-1.48
1.395	0.4275	-0.7258	-1.49
1.400	0.4251	-0.7264	-1.50
1.405	0.4226	-0.7269	-1.51
1.410	0.4201	-0.7274	-1.51
1.415	0.4177	-0.7280	-1.52
1.420	0.4152	-0.7284	-1.53
1.425	0.4126	-0.7290	-1.54
1.430	0.4101	-0.7295	-1.55
1.435	0.4074	-0.7300	-1.56
1.440	0.4050	-0.7307	-1.56
1.445	0.4024	-0.7312	-1.57
1.450	0.3999	-0.7317	-1.58
1.455	0.3973	-0.7323	-1.59
1.460	0.3948	-0.7327	-1.59
1.465	0.3920	-0.7331	-1.60
1.470	0.3894	-0.7335	-1.61
1.475	0.3868	-0.7340	-1.62
1.480	0.3841	-0.7346	-1.63
1.485	0.3814	-0.7350	-1.64
1.490	0.3786	-0.7356	-1.65
1.495	0.3760	-0.7359	-1.66

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.3733	-0.7364	-1.66
1.505	0.3706	-0.7366	-1.68
1.510	0.3676	-0.7371	-1.69
1.515	0.3648	-0.7375	-1.69
1.520	0.3621	-0.7379	-1.70
1.525	0.3592	-0.7383	-1.71
1.530	0.3564	-0.7386	-1.72
1.535	0.3535	-0.7392	-1.73
1.540	0.3506	-0.7395	-1.74
1.545	0.3478	-0.7398	-1.75
1.550	0.3450	-0.7400	-1.76
1.555	0.3419	-0.7403	-1.77
1.560	0.3388	-0.7406	-1.78
1.565	0.3358	-0.7409	-1.79
1.570	0.3329	-0.7412	-1.80
1.575	0.3300	-0.7417	-1.81
1.580	0.3271	-0.7421	-1.82
1.585	0.3240	-0.7423	-1.83
1.590	0.3211	-0.7428	-1.84
1.595	0.3182	-0.7430	-1.85
1.600	0.3153	-0.7436	-1.86
1.605	0.3125	-0.7436	-1.87
1.610	0.3095	-0.7438	-1.88
1.615	0.3067	-0.7440	-1.89
1.620	0.3037	-0.7442	-1.90
1.625	0.3007	-0.7444	-1.91
1.630	0.2977	-0.7445	-1.92
1.635	0.2948	-0.7447	-1.93
1.640	0.2917	-0.7448	-1.94
1.645	0.2889	-0.7448	-1.95
1.650	0.2859	-0.7451	-1.96
1.655	0.2831	-0.7453	-1.97
1.660	0.2802	-0.7453	-1.98
1.665	0.2772	-0.7454	-1.99
1.670	0.2746	-0.7454	-2.00
1.675	0.2717	-0.7455	-2.01
1.680	0.2690	-0.7456	-2.02
1.685	0.2662	-0.7457	-2.03
1.690	0.2636	-0.7456	-2.04
1.695	0.2610	-0.7456	-2.05
1.700	0.2583	-0.7455	-2.06
1.705	0.2558	-0.7455	-2.07
1.710	0.2529	-0.7456	-2.08
1.715	0.2502	-0.7454	-2.09
1.720	0.2474	-0.7454	-2.10
1.725	0.2448	-0.7454	-2.11

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.2422	-0.7454	-2.12
1.735	0.2398	-0.7451	-2.13
1.740	0.2373	-0.7453	-2.13
1.745	0.2348	-0.7454	-2.14
1.750	0.2326	-0.7453	-2.15
1.755	0.2303	-0.7453	-2.16
1.760	0.2281	-0.7453	-2.16
1.765	0.2259	-0.7454	-2.17
1.770	0.2239	-0.7454	-2.18
1.775	0.2220	-0.7452	-2.19
1.780	0.2199	-0.7449	-2.19
1.785	0.2180	-0.7447	-2.20
1.790	0.2158	-0.7445	-2.21
1.795	0.2136	-0.7443	-2.22
1.800	0.2115	-0.7443	-2.23
1.805	0.2094	-0.7444	-2.23
1.810	0.2075	-0.7443	-2.24
1.815	0.2057	-0.7445	-2.24
1.820	0.2038	-0.7444	-2.25
1.825	0.2021	-0.7443	-2.26
1.830	0.2005	-0.7441	-2.26
1.835	0.1987	-0.7440	-2.27
1.840	0.1969	-0.7438	-2.28
1.845	0.1953	-0.7437	-2.28
1.850	0.1937	-0.7434	-2.29
1.855	0.1920	-0.7431	-2.30
1.860	0.1901	-0.7429	-2.31
1.865	0.1887	-0.7427	-2.31
1.870	0.1870	-0.7429	-2.31
1.875	0.1852	-0.7427	-2.32
1.880	0.1837	-0.7430	-2.32
1.885	0.1823	-0.7431	-2.32
1.890	0.1811	-0.7431	-2.33
1.895	0.1798	-0.7431	-2.33
1.900	0.1783	-0.7428	-2.34
1.905	0.1769	-0.7427	-2.34
1.910	0.1755	-0.7429	-2.35
1.915	0.1739	-0.7428	-2.35
1.920	0.1725	-0.7430	-2.35
1.925	0.1712	-0.7431	-2.35
1.930	0.1699	-0.7431	-2.36
1.935	0.1687	-0.7431	-2.36
1.940	0.1671	-0.7431	-2.36
1.945	0.1660	-0.7431	-2.37
1.950	0.1647	-0.7431	-2.37
1.955	0.1635	-0.7431	-2.37

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.1622	-0.7431	-2.38
1.965	0.1608	-0.7431	-2.38
1.970	0.1597	-0.7430	-2.38
1.975	0.1585	-0.7430	-2.39
1.980	0.1572	-0.7430	-2.39
1.985	0.1561	-0.7429	-2.39
1.990	0.1548	-0.7427	-2.40
1.995	0.1535	-0.7428	-2.40
2.000	0.1521	-0.7426	-2.41
2.005	0.1507	-0.7426	-2.41
2.010	0.1494	-0.7426	-2.41
2.015	0.1481	-0.7425	-2.42
2.020	0.1468	-0.7425	-2.42
2.025	0.1454	-0.7424	-2.42
2.030	0.1439	-0.7424	-2.43
2.035	0.1425	-0.7424	-2.43
2.040	0.1411	-0.7422	-2.44
2.045	0.1395	-0.7421	-2.44
2.050	0.1381	-0.7420	-2.44
2.055	0.1367	-0.7419	-2.45
2.060	0.1351	-0.7417	-2.45
2.065	0.1335	-0.7417	-2.46
2.070	0.1318	-0.7416	-2.46
2.075	0.1302	-0.7415	-2.47
2.080	0.1286	-0.7412	-2.47
2.085	0.1269	-0.7411	-2.48
2.090	0.1250	-0.7408	-2.48
2.095	0.1233	-0.7407	-2.49
2.100	0.1215	-0.7406	-2.49
2.105	0.1195	-0.7404	-2.50
2.110	0.1176	-0.7404	-2.50
2.115	0.1157	-0.7401	-2.51
2.120	0.1136	-0.7401	-2.51
2.125	0.1116	-0.7400	-2.52
2.130	0.1096	-0.7396	-2.53
2.135	0.1075	-0.7396	-2.53
2.140	0.1055	-0.7393	-2.54
2.145	0.1034	-0.7392	-2.54
2.150	0.1013	-0.7389	-2.55
2.155	0.0992	-0.7386	-2.55
2.160	0.0971	-0.7384	-2.56
2.165	0.0949	-0.7382	-2.57
2.170	0.0927	-0.7379	-2.57
2.175	0.0905	-0.7376	-2.58
2.180	0.0883	-0.7374	-2.58
2.185	0.0861	-0.7370	-2.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.0838	-0.7367	-2.60
2.195	0.0815	-0.7365	-2.60
2.200	0.0792	-0.7362	-2.61
2.205	0.0770	-0.7358	-2.62
2.210	0.0745	-0.7354	-2.62
2.215	0.0723	-0.7350	-2.63
2.220	0.0700	-0.7349	-2.64
2.225	0.0677	-0.7345	-2.64
2.230	0.0653	-0.7341	-2.65
2.235	0.0629	-0.7337	-2.66
2.240	0.0606	-0.7336	-2.66
2.245	0.0585	-0.7331	-2.67
2.250	0.0560	-0.7329	-2.67
2.255	0.0538	-0.7325	-2.68
2.260	0.0516	-0.7322	-2.69
2.265	0.0492	-0.7318	-2.69
2.270	0.0469	-0.7313	-2.70
2.275	0.0447	-0.7310	-2.71
2.280	0.0423	-0.7306	-2.71
2.285	0.0402	-0.7302	-2.72
2.290	0.0380	-0.7298	-2.72
2.295	0.0358	-0.7294	-2.73
2.300	0.0337	-0.7291	-2.74
2.305	0.0313	-0.7284	-2.74
2.310	0.0291	-0.7280	-2.75
2.315	0.0270	-0.7277	-2.76
2.320	0.0248	-0.7272	-2.76
2.325	0.0227	-0.7268	-2.77
2.330	0.0207	-0.7264	-2.77
2.335	0.0185	-0.7260	-2.78
2.340	0.0164	-0.7255	-2.78
2.345	0.0143	-0.7250	-2.79
2.350	0.0124	-0.7246	-2.80
2.355	0.0105	-0.7240	-2.80
2.360	0.0087	-0.7236	-2.81
2.365	0.0067	-0.7231	-2.82
2.370	0.0050	-0.7227	-2.82
2.375	0.0031	-0.7222	-2.83
2.380	0.0016	-0.7217	-2.83
2.385	-0.0001	-0.7212	-2.84
2.390	-0.0019	-0.7209	-2.84
2.395	-0.0036	-0.7205	-2.85
2.400	-0.0053	-0.7202	-2.85
2.405	-0.0068	-0.7197	-2.86
2.410	-0.0082	-0.7193	-2.86
2.415	-0.0098	-0.7189	-2.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	-0.0113	-0.7184	-2.87
2.425	-0.0128	-0.7180	-2.88
2.430	-0.0141	-0.7176	-2.88
2.435	-0.0153	-0.7171	-2.89
2.440	-0.0166	-0.7167	-2.89
2.445	-0.0179	-0.7164	-2.89
2.450	-0.0190	-0.7160	-2.90
2.455	-0.0200	-0.7154	-2.91
2.460	-0.0211	-0.7150	-2.91
2.465	-0.0221	-0.7146	-2.91
2.470	-0.0232	-0.7142	-2.92
2.475	-0.0243	-0.7138	-2.92
2.480	-0.0253	-0.7135	-2.93
2.485	-0.0265	-0.7132	-2.93
2.490	-0.0273	-0.7128	-2.93
2.495	-0.0283	-0.7124	-2.94
2.500	-0.0292	-0.7121	-2.94
2.505	-0.0303	-0.7119	-2.94
2.510	-0.0313	-0.7117	-2.95
2.515	-0.0322	-0.7115	-2.95
2.520	-0.0331	-0.7112	-2.95
2.525	-0.0341	-0.7110	-2.95
2.530	-0.0351	-0.7109	-2.95
2.535	-0.0359	-0.7106	-2.96
2.540	-0.0370	-0.7104	-2.96
2.545	-0.0378	-0.7102	-2.96
2.550	-0.0386	-0.7100	-2.96
2.555	-0.0392	-0.7095	-2.97
2.560	-0.0399	-0.7091	-2.97
2.565	-0.0402	-0.7086	-2.98
2.570	-0.0407	-0.7081	-2.98
2.575	-0.0410	-0.7077	-2.99
2.580	-0.0413	-0.7072	-2.99
2.585	-0.0416	-0.7071	-3.00
2.590	-0.0418	-0.7069	-3.00
2.595	-0.0420	-0.7068	-3.00
2.600	-0.0423	-0.7068	-3.00
2.605	-0.0427	-0.7067	-3.00
2.610	-0.0430	-0.7065	-3.00
2.615	-0.0435	-0.7065	-3.00
2.620	-0.0438	-0.7062	-3.01
2.625	-0.0442	-0.7061	-3.01
2.630	-0.0446	-0.7059	-3.01
2.635	-0.0450	-0.7056	-3.01
2.640	-0.0454	-0.7053	-3.01
2.645	-0.0459	-0.7050	-3.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	-0.0463	-0.7048	-3.02
2.655	-0.0467	-0.7045	-3.02
2.660	-0.0472	-0.7041	-3.03
2.665	-0.0478	-0.7037	-3.03
2.670	-0.0482	-0.7034	-3.04
2.675	-0.0486	-0.7031	-3.04
2.680	-0.0491	-0.7030	-3.04
2.685	-0.0494	-0.7028	-3.04
2.690	-0.0499	-0.7027	-3.04
2.695	-0.0502	-0.7026	-3.04
2.700	-0.0505	-0.7024	-3.05
2.705	-0.0508	-0.7022	-3.05
2.710	-0.0512	-0.7020	-3.05
2.715	-0.0513	-0.7018	-3.05
2.720	-0.0519	-0.7015	-3.06
2.725	-0.0521	-0.7008	-3.06
2.730	-0.0528	-0.7003	-3.07
2.735	-0.0532	-0.6998	-3.08
2.740	-0.0539	-0.6989	-3.09
2.745	-0.0546	-0.6984	-3.09
2.750	-0.0553	-0.6979	-3.10
2.755	-0.0562	-0.6971	-3.11
2.760	-0.0572	-0.6964	-3.11
2.765	-0.0580	-0.6958	-3.12
2.770	-0.0587	-0.6952	-3.13
2.775	-0.0599	-0.6945	-3.13
2.780	-0.0607	-0.6939	-3.14
2.785	-0.0619	-0.6934	-3.15
2.790	-0.0629	-0.6928	-3.15
2.795	-0.0640	-0.6922	-3.16
2.800	-0.0651	-0.6917	-3.16
2.805	-0.0662	-0.6912	-3.17
2.810	-0.0672	-0.6905	-3.18
2.815	-0.0684	-0.6900	-3.18
2.820	-0.0696	-0.6897	-3.18
2.825	-0.0707	-0.6893	-3.19
2.830	-0.0715	-0.6888	-3.19
2.835	-0.0727	-0.6883	-3.20
2.840	-0.0738	-0.6878	-3.20
2.845	-0.0749	-0.6873	-3.21
2.850	-0.0758	-0.6868	-3.21
2.855	-0.0768	-0.6863	-3.22
2.860	-0.0778	-0.6859	-3.22
2.865	-0.0785	-0.6853	-3.23
2.870	-0.0793	-0.6847	-3.23
2.875	-0.0802	-0.6841	-3.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.880	-0.0809	-0.6832	-3.25
2.885	-0.0816	-0.6823	-3.26
2.890	-0.0827	-0.6813	-3.27
2.895	-0.0839	-0.6803	-3.28
2.900	-0.0853	-0.6793	-3.29
2.905	-0.0865	-0.6785	-3.30
2.910	-0.0877	-0.6778	-3.31
2.915	-0.0891	-0.6771	-3.31
2.920	-0.0907	-0.6765	-3.32
2.925	-0.0921	-0.6760	-3.32
2.930	-0.0935	-0.6753	-3.33
2.935	-0.0947	-0.6751	-3.33
2.940	-0.0959	-0.6749	-3.33
2.945	-0.0972	-0.6746	-3.33
2.950	-0.0982	-0.6745	-3.33
2.955	-0.0993	-0.6744	-3.33
2.960	-0.0999	-0.6745	-3.33
2.965	-0.1010	-0.6744	-3.33
2.970	-0.1014	-0.6744	-3.32
2.975	-0.1021	-0.6745	-3.32
2.980	-0.1028	-0.6745	-3.32
2.985	-0.1031	-0.6745	-3.32
2.990	-0.1033	-0.6743	-3.32
2.995	-0.1036	-0.6741	-3.32
3.000	-0.1040	-0.6738	-3.33
3.005	-0.1069	-0.6728	-3.33
3.010	-0.1073	-0.6728	-3.33
3.015	-0.1074	-0.6730	-3.33
3.020	-0.1075	-0.6728	-3.33
3.025	-0.1075	-0.6729	-3.33
3.030	-0.1075	-0.6727	-3.33
3.035	-0.1077	-0.6722	-3.34
3.040	-0.1074	-0.6716	-3.35
3.045	-0.1074	-0.6710	-3.36
3.050	-0.1070	-0.6701	-3.37
3.055	-0.1072	-0.6696	-3.37
3.060	-0.1076	-0.6686	-3.39
3.065	-0.1081	-0.6675	-3.40
3.070	-0.1086	-0.6666	-3.41
3.075	-0.1090	-0.6657	-3.42
3.080	-0.1098	-0.6647	-3.43
3.085	-0.1105	-0.6638	-3.44
3.090	-0.1117	-0.6632	-3.45
3.095	-0.1126	-0.6627	-3.45
3.100	-0.1138	-0.6625	-3.45
3.105	-0.1146	-0.6625	-3.45

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.110	-0.1159	-0.6622	-3.45
3.115	-0.1166	-0.6621	-3.45
3.120	-0.1179	-0.6621	-3.45
3.125	-0.1190	-0.6624	-3.44
3.130	-0.1197	-0.6628	-3.43
3.135	-0.1205	-0.6632	-3.43
3.140	-0.1209	-0.6639	-3.42
3.145	-0.1215	-0.6643	-3.41
3.150	-0.1220	-0.6650	-3.40
3.155	-0.1221	-0.6653	-3.40
3.160	-0.1223	-0.6658	-3.39
3.165	-0.1222	-0.6662	-3.38
3.170	-0.1223	-0.6663	-3.38
3.175	-0.1220	-0.6668	-3.38
3.180	-0.1217	-0.6667	-3.38
3.185	-0.1216	-0.6663	-3.38
3.190	-0.1215	-0.6660	-3.39
3.195	-0.1211	-0.6655	-3.40
3.200	-0.1212	-0.6648	-3.40
3.205	-0.1215	-0.6639	-3.41
3.210	-0.1216	-0.6630	-3.43
3.215	-0.1216	-0.6619	-3.44
3.220	-0.1224	-0.6610	-3.45
3.225	-0.1234	-0.6603	-3.46
3.230	-0.1244	-0.6599	-3.46
3.235	-0.1255	-0.6598	-3.46
3.240	-0.1265	-0.6599	-3.45
3.245	-0.1271	-0.6601	-3.45
3.250	-0.1281	-0.6602	-3.45
3.255	-0.1288	-0.6603	-3.44
3.260	-0.1297	-0.6608	-3.43
3.265	-0.1302	-0.6608	-3.43
3.270	-0.1307	-0.6609	-3.43
3.275	-0.1316	-0.6611	-3.43
3.280	-0.1321	-0.6611	-3.42
3.285	-0.1326	-0.6616	-3.42
3.290	-0.1334	-0.6615	-3.42
3.295	-0.1339	-0.6615	-3.41
3.300	-0.1344	-0.6614	-3.41
3.305	-0.1347	-0.6614	-3.41
3.310	-0.1353	-0.6611	-3.42
3.315	-0.1355	-0.6610	-3.42
3.320	-0.1360	-0.6610	-3.42
3.325	-0.1367	-0.6608	-3.42
3.330	-0.1366	-0.6602	-3.42
3.335	-0.1371	-0.6599	-3.43

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.340	-0.1374	-0.6596	-3.43
3.345	-0.1375	-0.6589	-3.44
3.350	-0.1380	-0.6581	-3.45
3.355	-0.1384	-0.6575	-3.45
3.360	-0.1388	-0.6566	-3.46
3.365	-0.1395	-0.6555	-3.48
3.370	-0.1398	-0.6544	-3.49
3.375	-0.1405	-0.6533	-3.50
3.380	-0.1411	-0.6525	-3.51
3.385	-0.1415	-0.6512	-3.53
3.390	-0.1424	-0.6501	-3.54
3.395	-0.1433	-0.6488	-3.55
3.400	-0.1441	-0.6475	-3.56
3.405	-0.1447	-0.6461	-3.58
3.410	-0.1458	-0.6448	-3.60
3.415	-0.1471	-0.6435	-3.61
3.420	-0.1483	-0.6426	-3.62
3.425	-0.1496	-0.6416	-3.63
3.430	-0.1509	-0.6403	-3.64
3.435	-0.1521	-0.6392	-3.65
3.440	-0.1535	-0.6379	-3.66
3.445	-0.1550	-0.6370	-3.67
3.450	-0.1567	-0.6362	-3.67
3.455	-0.1583	-0.6351	-3.68
3.460	-0.1596	-0.6343	-3.69
3.465	-0.1610	-0.6335	-3.69
3.470	-0.1626	-0.6333	-3.69
3.475	-0.1639	-0.6329	-3.69
3.480	-0.1652	-0.6324	-3.69
3.485	-0.1667	-0.6327	-3.68
3.490	-0.1679	-0.6329	-3.68
3.495	-0.1685	-0.6324	-3.68
3.500	-0.1692	-0.6324	-3.68
3.505	-0.1697	-0.6324	-3.68
3.510	-0.1701	-0.6320	-3.68
3.515	-0.1700	-0.6316	-3.69
3.520	-0.1701	-0.6309	-3.70
3.525	-0.1702	-0.6305	-3.70
3.530	-0.1700	-0.6294	-3.72
3.535	-0.1689	-0.6280	-3.74
3.540	-0.1686	-0.6262	-3.76
3.545	-0.1688	-0.6240	-3.79
3.550	-0.1691	-0.6213	-3.82
3.555	-0.1698	-0.6185	-3.86
3.560	-0.1710	-0.6158	-3.89
3.565	-0.1725	-0.6133	-3.92

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	-0.1748	-0.6107	-3.94
3.575	-0.1771	-0.6077	-3.97
3.580	-0.1808	-0.6065	-3.97
3.585	-0.1840	-0.6057	-3.97
3.590	-0.1875	-0.6065	-3.95
3.595	-0.1903	-0.6073	-3.93
3.600	-0.1921	-0.6090	-3.90
3.605	-0.1933	-0.6105	-3.87
3.610	-0.1936	-0.6122	-3.85
3.615	-0.1930	-0.6133	-3.84
3.620	-0.1923	-0.6141	-3.83
3.625	-0.1915	-0.6145	-3.83
3.630	-0.1897	-0.6144	-3.84
3.635	-0.1883	-0.6134	-3.85
3.640	-0.1877	-0.6119	-3.88
3.645	-0.1874	-0.6105	-3.90
3.650	-0.1874	-0.6090	-3.92
3.655	-0.1876	-0.6075	-3.93
3.660	-0.1879	-0.6060	-3.95
3.665	-0.1884	-0.6048	-3.97
3.670	-0.1888	-0.6032	-3.98
3.675	-0.1892	-0.6014	-4.01
3.680	-0.1896	-0.6001	-4.02
3.685	-0.1905	-0.5985	-4.04
3.690	-0.1916	-0.5968	-4.06
3.695	-0.1929	-0.5952	-4.07
3.700	-0.1943	-0.5932	-4.09
3.705	-0.1963	-0.5919	-4.10
3.710	-0.1987	-0.5901	-4.11
3.715	-0.2015	-0.5889	-4.12
3.720	-0.2047	-0.5880	-4.12
3.725	-0.2081	-0.5883	-4.10
3.730	-0.2125	-0.5885	-4.07
3.735	-0.2150	-0.5901	-4.04
3.740	-0.2178	-0.5924	-4.00
3.745	-0.2190	-0.5949	-3.96
3.750	-0.2198	-0.5982	-3.91
3.755	-0.2192	-0.6008	-3.88
3.760	-0.2178	-0.6030	-3.86
3.765	-0.2156	-0.6045	-3.85
3.770	-0.2127	-0.6048	-3.86
3.775	-0.2101	-0.6034	-3.89
3.780	-0.2070	-0.6016	-3.93
3.785	-0.2064	-0.5979	-3.98
3.790	-0.2072	-0.5945	-4.02
3.795	-0.2088	-0.5917	-4.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	-0.2108	-0.5896	-4.07
3.805	-0.2134	-0.5882	-4.07
3.810	-0.2157	-0.5875	-4.07
3.815	-0.2180	-0.5869	-4.07
3.820	-0.2202	-0.5865	-4.06
3.825	-0.2223	-0.5868	-4.05
3.830	-0.2244	-0.5869	-4.04
3.835	-0.2250	-0.5901	-3.99
3.840	-0.2262	-0.5897	-3.99
3.845	-0.2268	-0.5898	-3.99
3.850	-0.2278	-0.5894	-3.99
3.855	-0.2288	-0.5890	-3.99
3.860	-0.2298	-0.5887	-3.99
3.865	-0.2306	-0.5883	-3.99
3.870	-0.2317	-0.5877	-3.99
3.875	-0.2327	-0.5875	-3.99
3.880	-0.2340	-0.5869	-3.99
3.885	-0.2352	-0.5865	-3.99
3.890	-0.2365	-0.5860	-3.99
3.895	-0.2376	-0.5855	-3.99
3.900	-0.2391	-0.5850	-3.99
3.905	-0.2402	-0.5842	-3.99
3.910	-0.2417	-0.5837	-3.99
3.915	-0.2431	-0.5831	-3.99
3.920	-0.2446	-0.5826	-3.99
3.925	-0.2460	-0.5819	-3.99
3.930	-0.2472	-0.5814	-3.99
3.935	-0.2485	-0.5807	-3.99
3.940	-0.2500	-0.5801	-3.99
3.945	-0.2512	-0.5795	-3.99
3.950	-0.2527	-0.5785	-3.99
3.955	-0.2541	-0.5778	-4.00
3.960	-0.2556	-0.5771	-4.00
3.965	-0.2571	-0.5763	-4.00
3.970	-0.2583	-0.5753	-4.00
3.975	-0.2596	-0.5745	-4.01
3.980	-0.2612	-0.5735	-4.01
3.985	-0.2628	-0.5725	-4.01
3.990	-0.2640	-0.5715	-4.02
3.995	-0.2653	-0.5706	-4.02
4.000	-0.2669	-0.5698	-4.02
4.005	-0.2684	-0.5689	-4.03
4.010	-0.2698	-0.5678	-4.03
4.015	-0.2711	-0.5666	-4.04
4.020	-0.2725	-0.5657	-4.04
4.025	-0.2736	-0.5647	-4.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.030	-0.2751	-0.5638	-4.05
4.035	-0.2765	-0.5628	-4.05
4.040	-0.2779	-0.5617	-4.06
4.045	-0.2793	-0.5606	-4.06
4.050	-0.2804	-0.5596	-4.07
4.055	-0.2816	-0.5585	-4.08
4.060	-0.2828	-0.5575	-4.08
4.065	-0.2839	-0.5568	-4.08
4.070	-0.2854	-0.5554	-4.09
4.075	-0.2863	-0.5546	-4.09
4.080	-0.2873	-0.5538	-4.10
4.085	-0.2886	-0.5527	-4.10
4.090	-0.2897	-0.5517	-4.11
4.095	-0.2908	-0.5510	-4.11
4.100	-0.2918	-0.5502	-4.11
4.105	-0.2930	-0.5493	-4.12
4.110	-0.2942	-0.5487	-4.12
4.115	-0.2954	-0.5481	-4.12
4.120	-0.2965	-0.5475	-4.11
4.125	-0.2979	-0.5469	-4.11
4.130	-0.2994	-0.5460	-4.11
4.135	-0.3006	-0.5454	-4.11
4.140	-0.3017	-0.5446	-4.12
4.145	-0.3030	-0.5432	-4.12
4.150	-0.3041	-0.5424	-4.13
4.155	-0.3050	-0.5416	-4.13
4.160	-0.3058	-0.5405	-4.14
4.165	-0.3064	-0.5398	-4.14
4.170	-0.3069	-0.5388	-4.15
4.175	-0.3074	-0.5383	-4.15
4.180	-0.3077	-0.5376	-4.16
4.185	-0.3079	-0.5369	-4.17
4.190	-0.3081	-0.5367	-4.17
4.195	-0.3084	-0.5363	-4.17
4.200	-0.3086	-0.5358	-4.18
4.205	-0.3090	-0.5360	-4.17
4.210	-0.3090	-0.5355	-4.18
4.215	-0.3091	-0.5354	-4.18
4.220	-0.3090	-0.5354	-4.18
4.225	-0.3096	-0.5353	-4.17
4.230	-0.3093	-0.5355	-4.18
4.235	-0.3093	-0.5354	-4.18
4.240	-0.3093	-0.5354	-4.18
4.245	-0.3094	-0.5357	-4.17
4.250	-0.3092	-0.5360	-4.17
4.255	-0.3093	-0.5362	-4.17

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.3092	-0.5362	-4.17
4.265	-0.3092	-0.5365	-4.16
4.270	-0.3091	-0.5368	-4.16
4.275	-0.3089	-0.5372	-4.16
4.280	-0.3089	-0.5374	-4.16
4.285	-0.3087	-0.5377	-4.15
4.290	-0.3085	-0.5378	-4.15
4.295	-0.3085	-0.5381	-4.15
4.300	-0.3083	-0.5385	-4.14
4.305	-0.3083	-0.5388	-4.14
4.310	-0.3082	-0.5391	-4.14
4.315	-0.3081	-0.5394	-4.14
4.320	-0.3081	-0.5398	-4.13
4.325	-0.3081	-0.5403	-4.13
4.330	-0.3079	-0.5405	-4.12
4.335	-0.3079	-0.5411	-4.12
4.340	-0.3079	-0.5416	-4.11
4.345	-0.3077	-0.5423	-4.10
4.350	-0.3077	-0.5430	-4.09
4.355	-0.3075	-0.5435	-4.09
4.360	-0.3071	-0.5442	-4.08
4.365	-0.3068	-0.5449	-4.08
4.370	-0.3066	-0.5455	-4.07
4.375	-0.3063	-0.5462	-4.07
4.380	-0.3058	-0.5468	-4.06
4.385	-0.3053	-0.5476	-4.05
4.390	-0.3047	-0.5482	-4.05
4.395	-0.3041	-0.5487	-4.05
4.400	-0.3037	-0.5492	-4.05
4.405	-0.3030	-0.5496	-4.05
4.410	-0.3026	-0.5499	-4.05
4.415	-0.3021	-0.5500	-4.05
4.420	-0.3017	-0.5503	-4.05
4.425	-0.3009	-0.5507	-4.05
4.430	-0.3006	-0.5507	-4.05
4.435	-0.3005	-0.5507	-4.05
4.440	-0.3004	-0.5507	-4.05
4.445	-0.3001	-0.5508	-4.05
4.450	-0.3003	-0.5511	-4.05
4.455	-0.2997	-0.5515	-4.04
4.460	-0.2996	-0.5519	-4.04
4.465	-0.2994	-0.5525	-4.04
4.470	-0.2984	-0.5531	-4.03
4.475	-0.2984	-0.5538	-4.03
4.480	-0.2978	-0.5550	-4.02
4.485	-0.2973	-0.5562	-4.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	-0.2970	-0.5571	-3.99
4.495	-0.2968	-0.5581	-3.98
4.500	-0.2962	-0.5591	-3.98
4.505	-0.2956	-0.5597	-3.97
4.510	-0.2952	-0.5602	-3.97
4.515	-0.2945	-0.5607	-3.97
4.520	-0.2939	-0.5609	-3.97
4.525	-0.2932	-0.5606	-3.98
4.530	-0.2928	-0.5603	-3.98
4.535	-0.2924	-0.5599	-3.99
4.540	-0.2925	-0.5596	-3.99
4.545	-0.2923	-0.5589	-4.00
4.550	-0.2926	-0.5588	-4.00
4.555	-0.2928	-0.5583	-4.01
4.560	-0.2926	-0.5580	-4.01
4.565	-0.2928	-0.5573	-4.02
4.570	-0.2929	-0.5568	-4.03
4.575	-0.2930	-0.5565	-4.03
4.580	-0.2931	-0.5558	-4.04
4.585	-0.2934	-0.5552	-4.04
4.590	-0.2933	-0.5545	-4.05
4.595	-0.2938	-0.5538	-4.06
4.600	-0.2941	-0.5532	-4.06
4.605	-0.2944	-0.5524	-4.07
4.610	-0.2947	-0.5520	-4.07
4.615	-0.2952	-0.5514	-4.08
4.620	-0.2953	-0.5505	-4.09
4.625	-0.2958	-0.5499	-4.09
4.630	-0.2959	-0.5495	-4.10
4.635	-0.2959	-0.5486	-4.11
4.640	-0.2962	-0.5480	-4.11
4.645	-0.2963	-0.5474	-4.12
4.650	-0.2964	-0.5468	-4.12
4.655	-0.2967	-0.5461	-4.13
4.660	-0.2969	-0.5457	-4.14
4.665	-0.2968	-0.5450	-4.14
4.670	-0.2967	-0.5444	-4.15
4.675	-0.2964	-0.5440	-4.16
4.680	-0.2964	-0.5434	-4.17
4.685	-0.2962	-0.5430	-4.17
4.690	-0.2962	-0.5423	-4.18
4.695	-0.2959	-0.5421	-4.19
4.700	-0.2958	-0.5416	-4.19
4.705	-0.2958	-0.5408	-4.20
4.710	-0.2954	-0.5401	-4.21
4.715	-0.2956	-0.5394	-4.22

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	-0.2959	-0.5386	-4.23
4.725	-0.2963	-0.5377	-4.24
4.730	-0.2966	-0.5367	-4.25
4.735	-0.2971	-0.5357	-4.26
4.740	-0.2978	-0.5348	-4.26
4.745	-0.2986	-0.5338	-4.27
4.750	-0.2994	-0.5328	-4.28
4.755	-0.3002	-0.5319	-4.28
4.760	-0.3011	-0.5309	-4.29
4.765	-0.3016	-0.5302	-4.29
4.770	-0.3021	-0.5296	-4.30
4.775	-0.3023	-0.5290	-4.30
4.780	-0.3025	-0.5281	-4.31
4.785	-0.3026	-0.5277	-4.32
4.790	-0.3021	-0.5274	-4.32
4.795	-0.3019	-0.5266	-4.34
4.800	-0.3014	-0.5258	-4.35
4.805	-0.3009	-0.5257	-4.35
4.810	-0.3001	-0.5249	-4.37
4.815	-0.3001	-0.5243	-4.38
4.820	-0.2993	-0.5234	-4.39
4.825	-0.2988	-0.5231	-4.40
4.830	-0.2987	-0.5226	-4.41
4.835	-0.2984	-0.5222	-4.42
4.840	-0.2981	-0.5222	-4.42
4.845	-0.2980	-0.5222	-4.42
4.850	-0.2982	-0.5220	-4.42
4.855	-0.2984	-0.5222	-4.42
4.860	-0.2983	-0.5220	-4.42
4.865	-0.2989	-0.5222	-4.41
4.870	-0.2990	-0.5226	-4.41
4.875	-0.2990	-0.5228	-4.40
4.880	-0.2994	-0.5230	-4.40
4.885	-0.2996	-0.5233	-4.39
4.890	-0.2998	-0.5236	-4.39
4.895	-0.3000	-0.5240	-4.38
4.900	-0.3003	-0.5244	-4.37
4.905	-0.3006	-0.5244	-4.37
4.910	-0.3010	-0.5248	-4.37
4.915	-0.3013	-0.5251	-4.36
4.920	-0.3018	-0.5254	-4.35
4.925	-0.3021	-0.5258	-4.35
4.930	-0.3023	-0.5260	-4.34
4.935	-0.3027	-0.5264	-4.33
4.940	-0.3030	-0.5265	-4.33
4.945	-0.3033	-0.5269	-4.32

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	-0.3037	-0.5271	-4.32
4.955	-0.3041	-0.5273	-4.31
4.960	-0.3044	-0.5275	-4.31
4.965	-0.3047	-0.5278	-4.30
4.970	-0.3052	-0.5280	-4.30
4.975	-0.3055	-0.5282	-4.29
4.980	-0.3060	-0.5283	-4.29
4.985	-0.3062	-0.5286	-4.28
4.990	-0.3068	-0.5286	-4.28
4.995	-0.3070	-0.5288	-4.27
5.000	-0.3075	-0.5288	-4.27
5.005	-0.3077	-0.5289	-4.27
5.010	-0.3082	-0.5287	-4.26
5.015	-0.3082	-0.5286	-4.27
5.020	-0.3084	-0.5285	-4.27
5.025	-0.3084	-0.5285	-4.27
5.030	-0.3084	-0.5280	-4.27
5.035	-0.3079	-0.5281	-4.28
5.040	-0.3076	-0.5276	-4.28
5.045	-0.3069	-0.5279	-4.28
5.050	-0.3067	-0.5277	-4.29
5.055	-0.3062	-0.5280	-4.29
5.060	-0.3059	-0.5277	-4.29
5.065	-0.3060	-0.5283	-4.29
5.070	-0.3053	-0.5281	-4.29
5.075	-0.3057	-0.5285	-4.29
5.080	-0.3054	-0.5283	-4.29
5.085	-0.3061	-0.5284	-4.28
5.090	-0.3059	-0.5285	-4.28
5.095	-0.3068	-0.5280	-4.28
5.100	-0.3071	-0.5280	-4.28
5.105	-0.3076	-0.5278	-4.28
5.110	-0.3083	-0.5275	-4.28
5.115	-0.3089	-0.5271	-4.28
5.120	-0.3094	-0.5269	-4.28
5.125	-0.3099	-0.5263	-4.28
5.130	-0.3107	-0.5259	-4.28
5.135	-0.3110	-0.5251	-4.29
5.140	-0.3120	-0.5246	-4.29
5.145	-0.3128	-0.5238	-4.29
5.150	-0.3132	-0.5232	-4.30
5.155	-0.3139	-0.5223	-4.30
5.160	-0.3145	-0.5215	-4.31
5.165	-0.3153	-0.5205	-4.31
5.170	-0.3162	-0.5195	-4.32
5.175	-0.3168	-0.5185	-4.33

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.180	-0.3176	-0.5173	-4.34
5.185	-0.3184	-0.5160	-4.35
5.190	-0.3193	-0.5149	-4.35
5.195	-0.3201	-0.5136	-4.36
5.200	-0.3210	-0.5121	-4.37
5.205	-0.3216	-0.5107	-4.39
5.210	-0.3225	-0.5094	-4.39
5.215	-0.3235	-0.5079	-4.40
5.220	-0.3242	-0.5063	-4.42
5.225	-0.3247	-0.5049	-4.43
5.230	-0.3256	-0.5031	-4.45
5.235	-0.3264	-0.5018	-4.46
5.240	-0.3272	-0.5000	-4.47
5.245	-0.3281	-0.4986	-4.48
5.250	-0.3289	-0.4968	-4.50
5.255	-0.3298	-0.4952	-4.51
5.260	-0.3306	-0.4939	-4.52
5.265	-0.3318	-0.4922	-4.53
5.270	-0.3326	-0.4908	-4.54
5.275	-0.3334	-0.4895	-4.55
5.280	-0.3343	-0.4879	-4.56
5.285	-0.3351	-0.4866	-4.57
5.290	-0.3359	-0.4854	-4.58
5.295	-0.3367	-0.4842	-4.59
5.300	-0.3376	-0.4829	-4.59
5.305	-0.3385	-0.4817	-4.60
5.310	-0.3392	-0.4804	-4.61
5.315	-0.3399	-0.4792	-4.62
5.320	-0.3407	-0.4780	-4.63
5.325	-0.3414	-0.4768	-4.64
5.330	-0.3419	-0.4757	-4.65
5.335	-0.3426	-0.4745	-4.65
5.340	-0.3432	-0.4732	-4.66
5.345	-0.3439	-0.4723	-4.67
5.350	-0.3445	-0.4707	-4.68
5.355	-0.3451	-0.4695	-4.69
5.360	-0.3457	-0.4682	-4.70
5.365	-0.3464	-0.4668	-4.71
5.370	-0.3469	-0.4656	-4.72
5.375	-0.3479	-0.4642	-4.73
5.380	-0.3486	-0.4629	-4.74
5.385	-0.3499	-0.4617	-4.74
5.390	-0.3506	-0.4605	-4.75
5.395	-0.3515	-0.4594	-4.76
5.400	-0.3524	-0.4584	-4.76
5.405	-0.3533	-0.4575	-4.76

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	-0.3545	-0.4566	-4.76
5.415	-0.3555	-0.4561	-4.76
5.420	-0.3562	-0.4549	-4.76
5.425	-0.3573	-0.4540	-4.77
5.430	-0.3579	-0.4530	-4.77
5.435	-0.3589	-0.4525	-4.77
5.440	-0.3596	-0.4518	-4.77
5.445	-0.3604	-0.4511	-4.77
5.450	-0.3612	-0.4504	-4.77
5.455	-0.3618	-0.4500	-4.77
5.460	-0.3624	-0.4494	-4.77
5.465	-0.3632	-0.4486	-4.77
5.470	-0.3638	-0.4481	-4.77
5.475	-0.3643	-0.4474	-4.78
5.480	-0.3652	-0.4466	-4.78
5.485	-0.3660	-0.4458	-4.78
5.490	-0.3667	-0.4454	-4.78
5.495	-0.3677	-0.4449	-4.77
5.500	-0.3685	-0.4444	-4.77
5.505	-0.3693	-0.4440	-4.77
5.510	-0.3705	-0.4437	-4.76
5.515	-0.3714	-0.4436	-4.75
5.520	-0.3722	-0.4434	-4.75
5.525	-0.3729	-0.4436	-4.74
5.530	-0.3740	-0.4436	-4.73
5.535	-0.3745	-0.4437	-4.72
5.540	-0.3753	-0.4438	-4.71
5.545	-0.3758	-0.4440	-4.71
5.550	-0.3766	-0.4440	-4.70
5.555	-0.3772	-0.4440	-4.69
5.560	-0.3779	-0.4440	-4.69
5.565	-0.3786	-0.4442	-4.68
5.570	-0.3793	-0.4441	-4.67
5.575	-0.3800	-0.4442	-4.66
5.580	-0.3808	-0.4443	-4.66
5.585	-0.3815	-0.4443	-4.65
5.590	-0.3823	-0.4443	-4.64
5.595	-0.3832	-0.4444	-4.63
5.600	-0.3841	-0.4445	-4.62
5.605	-0.3851	-0.4449	-4.61
5.610	-0.3859	-0.4451	-4.60
5.615	-0.3868	-0.4456	-4.58
5.620	-0.3875	-0.4459	-4.57
5.625	-0.3883	-0.4462	-4.56
5.630	-0.3889	-0.4467	-4.55
5.635	-0.3895	-0.4472	-4.54

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.3901	-0.4474	-4.53
5.645	-0.3906	-0.4478	-4.52
5.650	-0.3911	-0.4482	-4.51
5.655	-0.3916	-0.4483	-4.51
5.660	-0.3921	-0.4484	-4.50
5.665	-0.3925	-0.4487	-4.49
5.670	-0.3931	-0.4491	-4.48
5.675	-0.3934	-0.4493	-4.48
5.680	-0.3941	-0.4496	-4.47
5.685	-0.3941	-0.4495	-4.47
5.690	-0.3944	-0.4496	-4.47
5.695	-0.3948	-0.4495	-4.46
5.700	-0.3951	-0.4496	-4.46
5.705	-0.3952	-0.4498	-4.46
5.710	-0.3955	-0.4497	-4.45
5.715	-0.3959	-0.4497	-4.45
5.720	-0.3960	-0.4497	-4.45
5.725	-0.3964	-0.4498	-4.44
5.730	-0.3964	-0.4497	-4.44
5.735	-0.3968	-0.4497	-4.44
5.740	-0.3970	-0.4494	-4.44
5.745	-0.3971	-0.4498	-4.44
5.750	-0.3972	-0.4491	-4.44
5.755	-0.3970	-0.4489	-4.45
5.760	-0.3971	-0.4488	-4.45
5.765	-0.3974	-0.4485	-4.45
5.770	-0.3970	-0.4479	-4.46
5.775	-0.3972	-0.4473	-4.46
5.780	-0.3976	-0.4468	-4.46
5.785	-0.3974	-0.4464	-4.47
5.790	-0.3973	-0.4457	-4.48
5.795	-0.3975	-0.4448	-4.49
5.800	-0.3974	-0.4442	-4.49
5.805	-0.3974	-0.4435	-4.50
5.810	-0.3974	-0.4425	-4.51
5.815	-0.3976	-0.4418	-4.52
5.820	-0.3975	-0.4406	-4.53
5.825	-0.3977	-0.4400	-4.54
5.830	-0.3976	-0.4389	-4.55
5.835	-0.3975	-0.4381	-4.56
5.840	-0.3977	-0.4371	-4.57
5.845	-0.3978	-0.4360	-4.58
5.850	-0.3979	-0.4353	-4.59
5.855	-0.3981	-0.4342	-4.60
5.860	-0.3982	-0.4331	-4.61
5.865	-0.3985	-0.4320	-4.62

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.3988	-0.4310	-4.62
5.875	-0.3991	-0.4299	-4.63
5.880	-0.3996	-0.4289	-4.64
5.885	-0.3997	-0.4282	-4.65
5.890	-0.4002	-0.4274	-4.65
5.895	-0.4002	-0.4264	-4.66
5.900	-0.4005	-0.4256	-4.67
5.905	-0.4008	-0.4250	-4.67
5.910	-0.4011	-0.4244	-4.67
5.915	-0.4012	-0.4239	-4.68
5.920	-0.4012	-0.4230	-4.69
5.925	-0.4014	-0.4221	-4.69
5.930	-0.4016	-0.4216	-4.70
5.935	-0.4015	-0.4208	-4.71
5.940	-0.4018	-0.4197	-4.72
5.945	-0.4020	-0.4188	-4.72
5.950	-0.4021	-0.4181	-4.73
5.955	-0.4024	-0.4169	-4.74
5.960	-0.4027	-0.4161	-4.75
5.965	-0.4032	-0.4154	-4.75
5.970	-0.4040	-0.4146	-4.75
5.975	-0.4047	-0.4141	-4.75
5.980	-0.4058	-0.4138	-4.74
5.985	-0.4062	-0.4138	-4.73
5.990	-0.4066	-0.4136	-4.73
5.995	-0.4073	-0.4136	-4.72
6.000	-0.4076	-0.4135	-4.72

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.